

Chapter 45

Hormones and the Endocrine System

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

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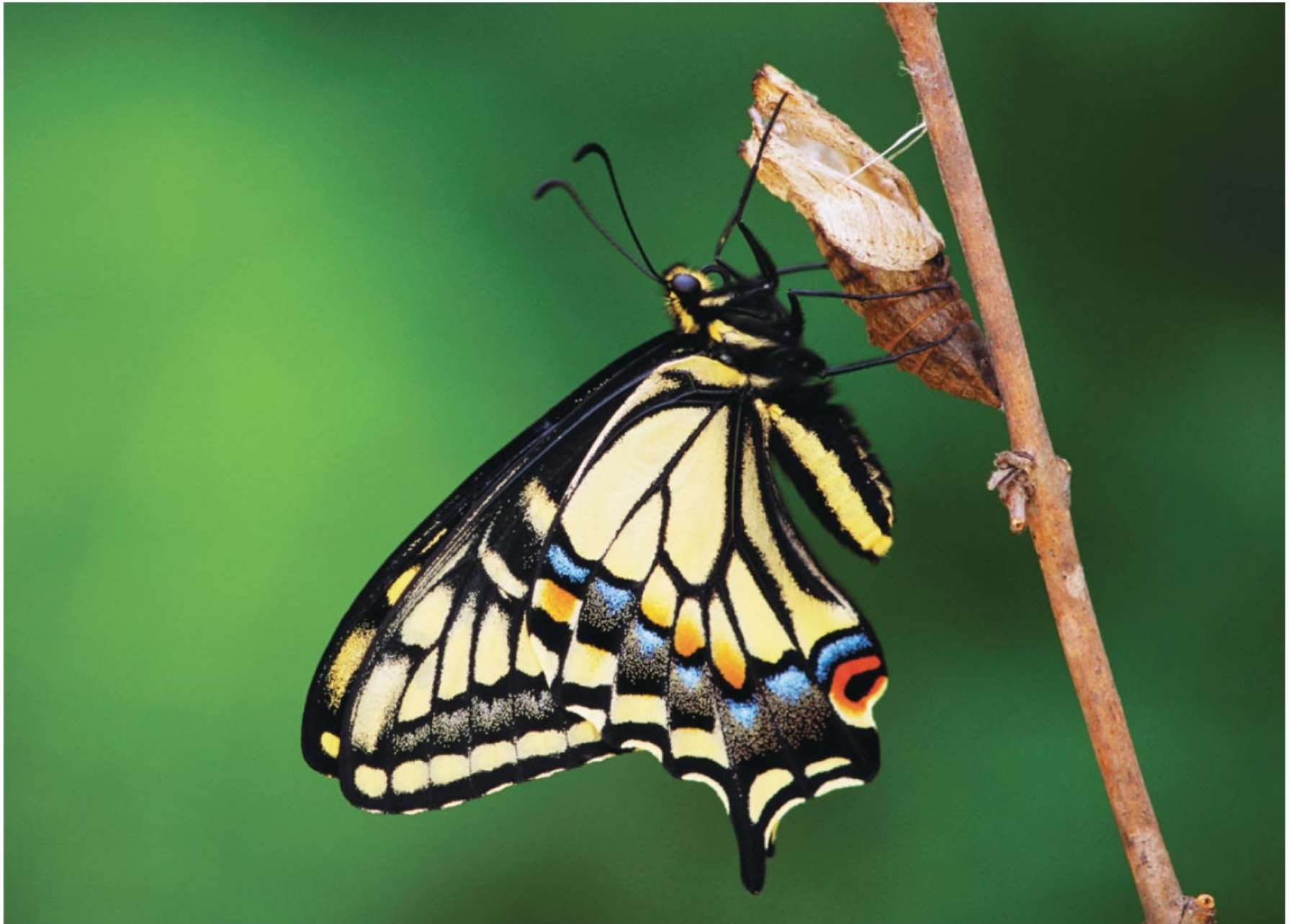
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Overview: The Body's Long-Distance Regulators

- Animal **hormones** are **chemical signals** that are **secreted** into the **circulatory system** and communicate **regulatory messages** within the body.
- Hormones reach all parts of the body, but only **target cells** are equipped to **respond**.
- Insect metamorphosis is regulated by hormones.

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- Two systems **coordinate communication** throughout the body: the endocrine system and the nervous system.
 - The **endocrine system** secretes hormones that coordinate **slower but longer-acting** responses including reproduction, development, energy metabolism, growth, and behavior.
 - The **nervous system** conveys **high-speed** electrical signals along specialized cells called neurons; these signals regulate other cells.

What role do hormones play in transforming a caterpillar into a butterfly?



Hormones and other signaling molecules bind to target receptors, triggering specific response pathways

- **Chemical signals bind to receptor proteins on target cells.**
- Only target cells respond to the signal.

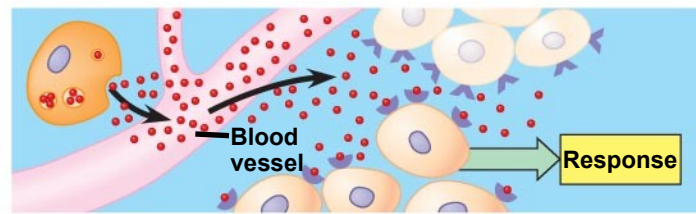
Types of Secreted Signaling Molecules

- **Secreted chemical signals** include
 - *Hormones*
 - *Local regulators*
 - *Neurotransmitters*
 - *Neurohormones*
 - *Pheromones*

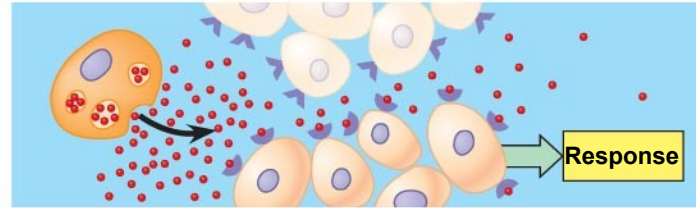
Hormones

- Endocrine signals (hormones) are secreted into extracellular fluids and travel via the bloodstream.
- **Endocrine glands** are **ductless** and **secrete hormones** directly into surrounding fluid.
- ***Hormones mediate responses to environmental stimuli and regulate growth, development, and reproduction.***
- **Exocrine glands** have **ducts** and secrete substances onto body surfaces or into body cavities (for example, tear ducts).

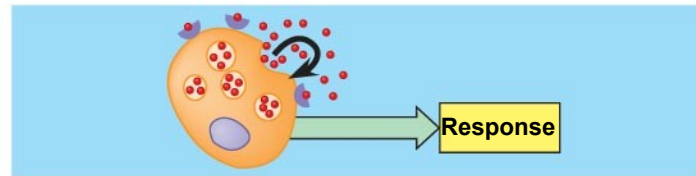
Intercellular communication by secreted molecules



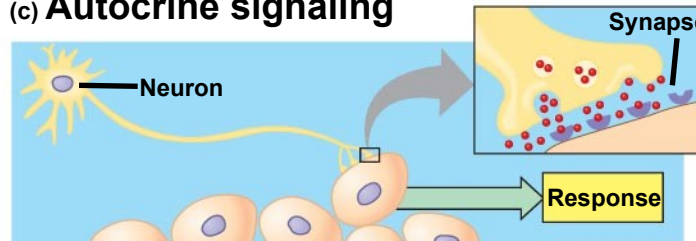
(a) **Endocrine signaling**



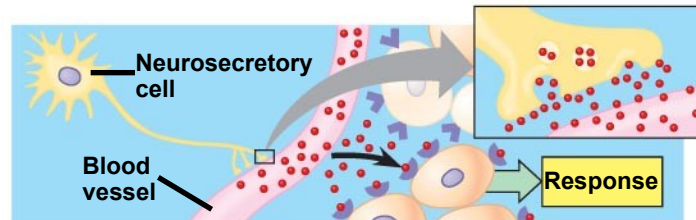
(b) **Paracrine signaling**



(c) **Autocrine signaling**



(d) **Synaptic signaling**

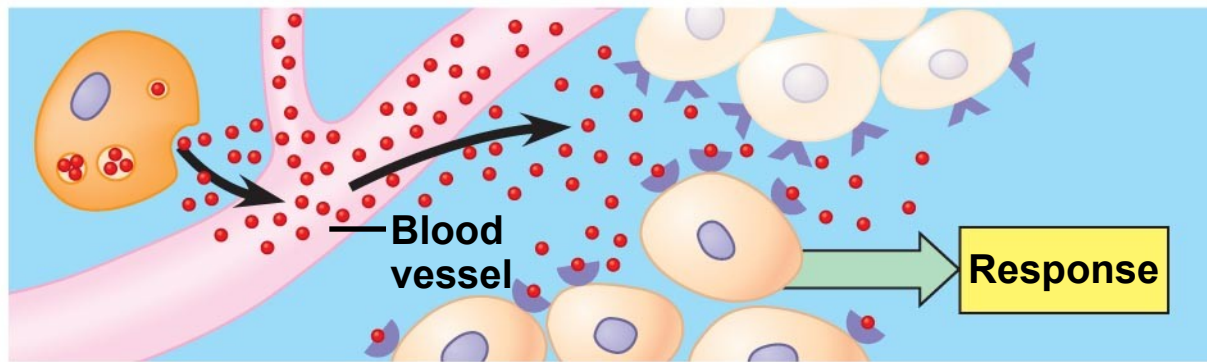


(e) **Neuroendocrine signaling**

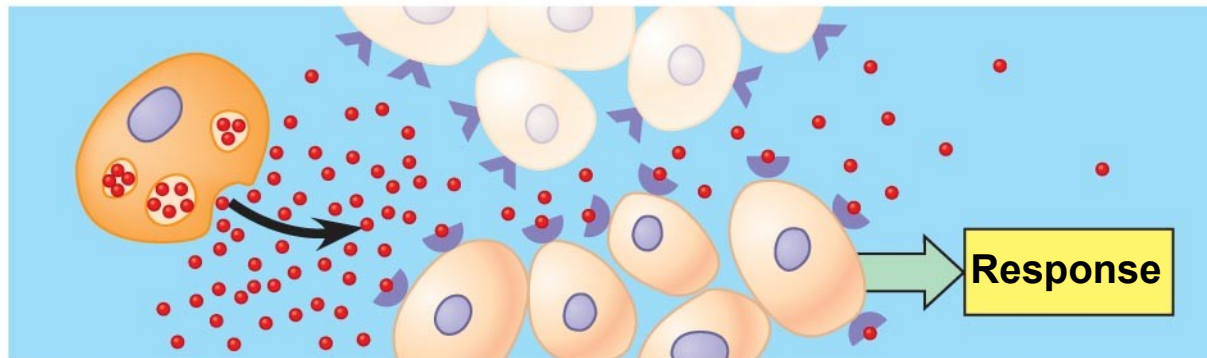
Local Regulators = Short Distance Chemical Signals

- **Local regulators** are chemical signals that travel over **short distances** by diffusion.
- Local regulators help regulate blood pressure, nervous system function, and reproduction.
- Local regulators are divided into two types:
 - **Paracrine** signals act on cells near the secreting cell.
 - **Autocrine** signals act on the secreting cell itself.

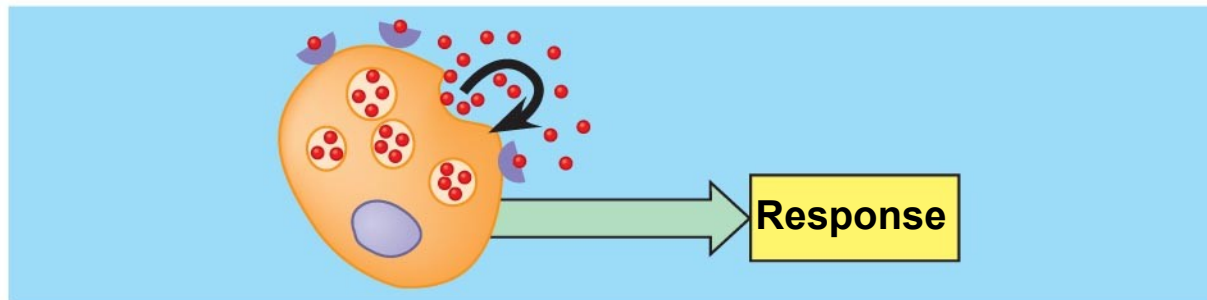
Intercellular
communication
by secreted
molecules



(a) **Endocrine** signaling



(b) **Paracrine** signaling

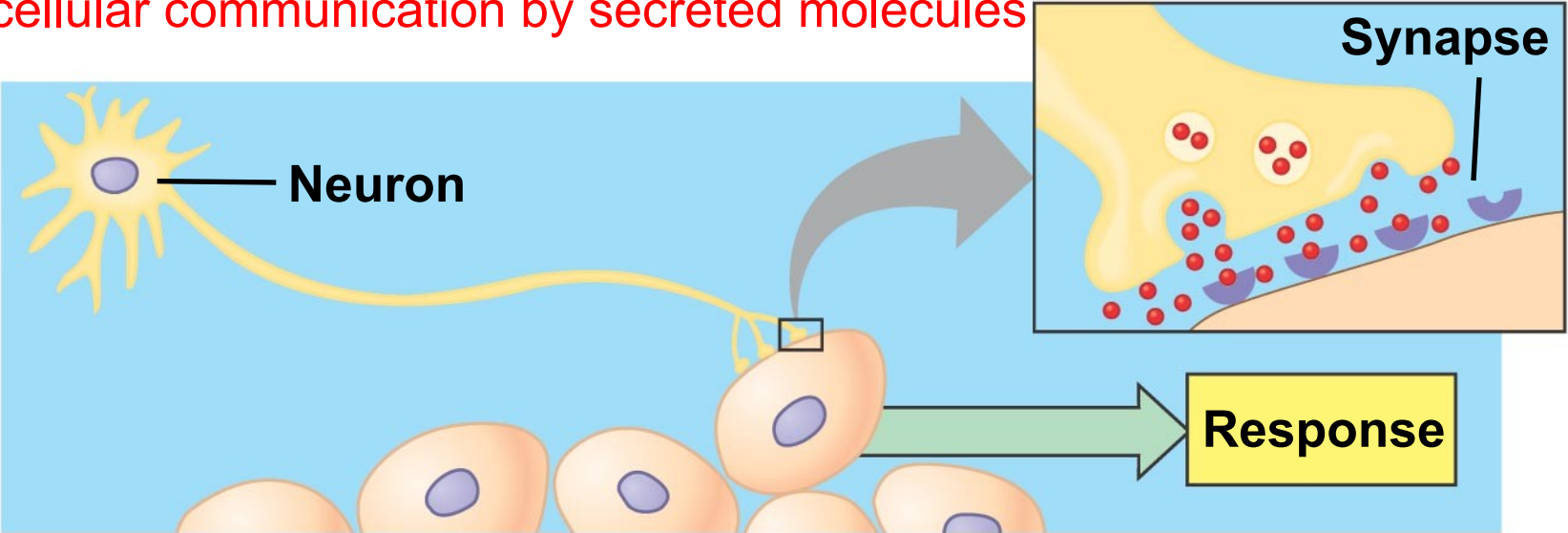


(c) **Autocrine** signaling

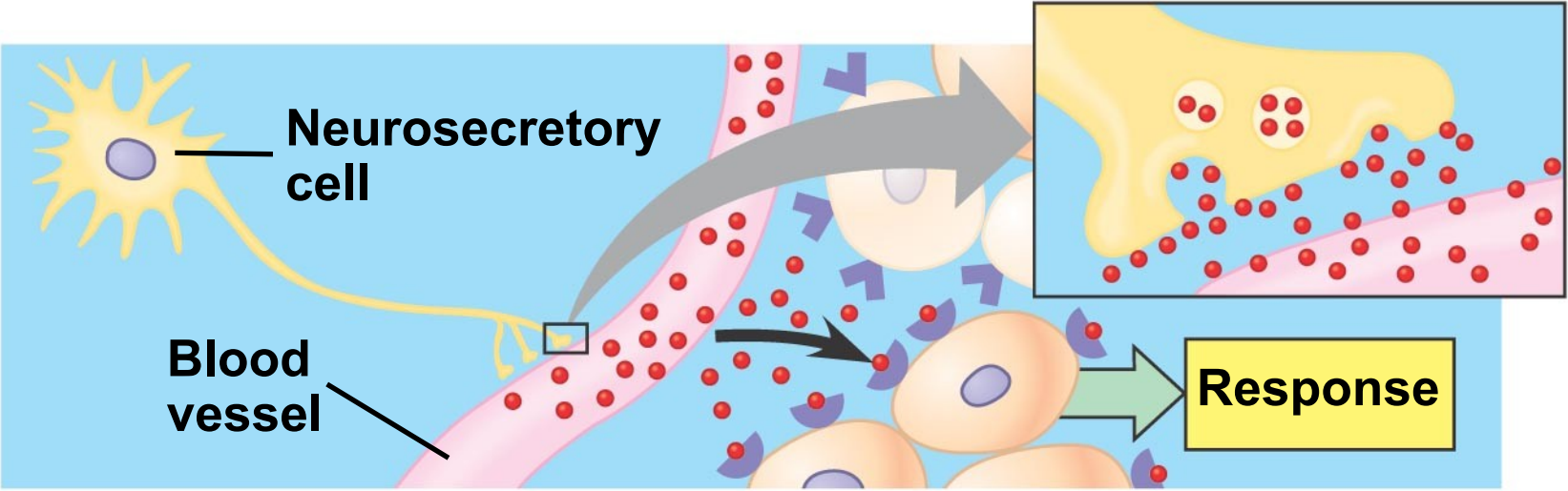
Neurotransmitters and Neurohormones

- **Neurons** (nerve cells) **contact target cells at synapses.**
- **At synapses**, neurons often **secrete chemical signals** called **neurotransmitters** that diffuse a short distance to bind to receptors on the target cell. Neurotransmitters play a role in sensation, memory, cognition, and movement.
- **Neurohormones** are a class of hormones that originate **from neurons in the brain** and diffuse through the **bloodstream.**

Intercellular communication by secreted molecules



(d) Synaptic signaling - neurotransmitters



(e) Neuroendocrine signaling

Pheromones

- ***Pheromones*** are chemical signals that are released from the body and used to *communicate with other individuals in the species*.
- Pheromones mark trails to food sources, warn of predators, and attract potential mates.

Chemical Classes of Hormones

- Three major classes of molecules function as hormones in vertebrates:
 - Polypeptides (proteins and peptides)
 - Amines derived from amino acids
 - Steroid hormones

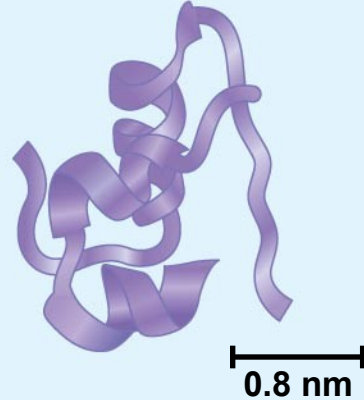
Polypeptides and amines are water-soluble.

Steroids are lipid-soluble.

-
- **Lipid-soluble hormones** (**steroid** hormones) pass easily through cell membranes.
 - **Water-soluble hormones** (**polypeptides** and **amines**) do not pass through the cell membrane.
 - ***The solubility of a hormone correlates with the location of receptors inside **or** on the surface of target cells.***

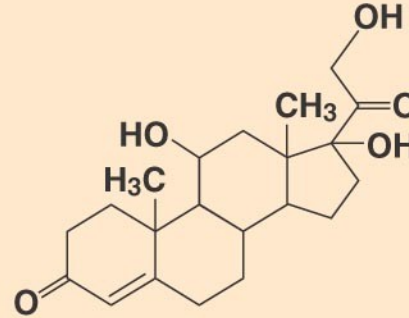
Hormones differ in form and solubility

Water-soluble

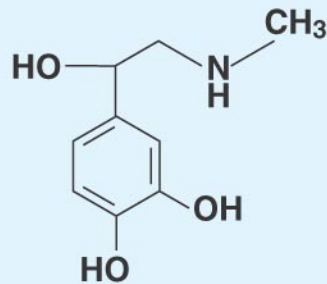


**Polypeptide:
Insulin**

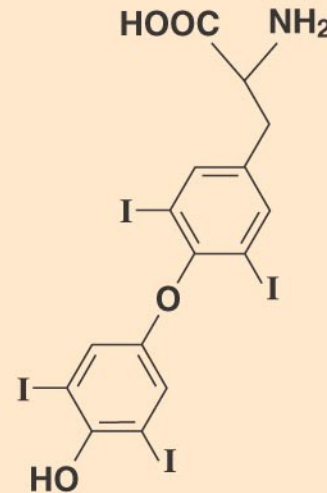
Lipid-soluble



**Steroid:
Cortisol**



**Amine:
Epinephrine**



**Amine:
Thyroxine**

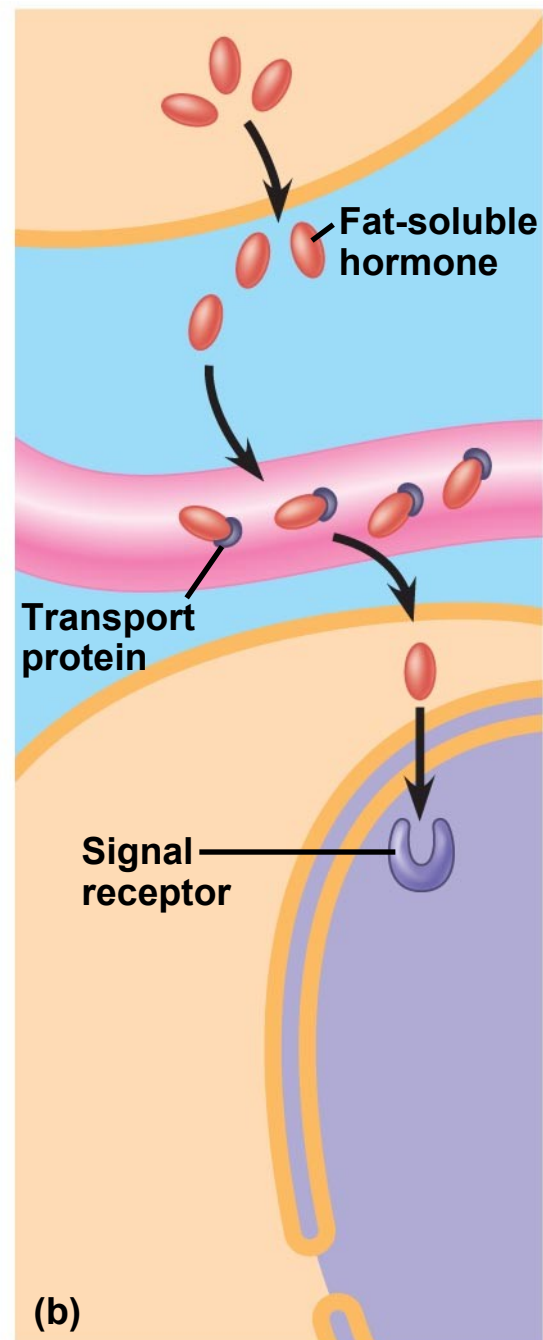
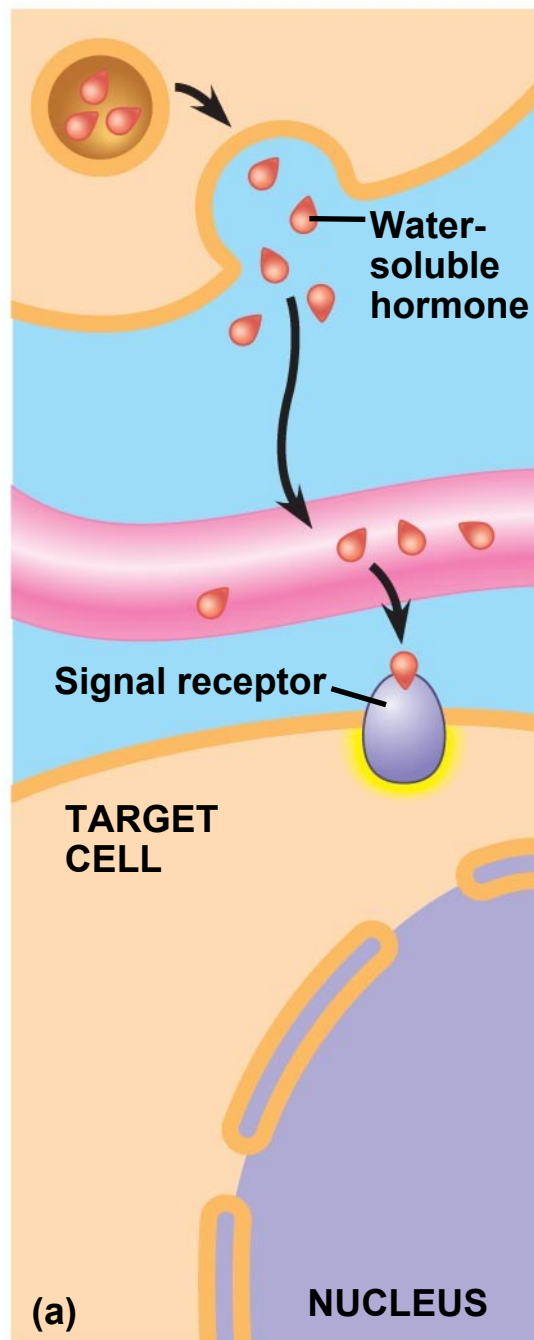
Cellular Response Pathways

- Water and lipid soluble hormones differ in their paths through a body.
- *Water-soluble hormones* are secreted by exocytosis, travel freely in the bloodstream, and *bind to cell-surface receptors*.
- *Lipid-soluble hormones* diffuse across cell membranes, travel in the bloodstream *bound to transport proteins*, and *diffuse through* the plasma *membrane* of target cells.

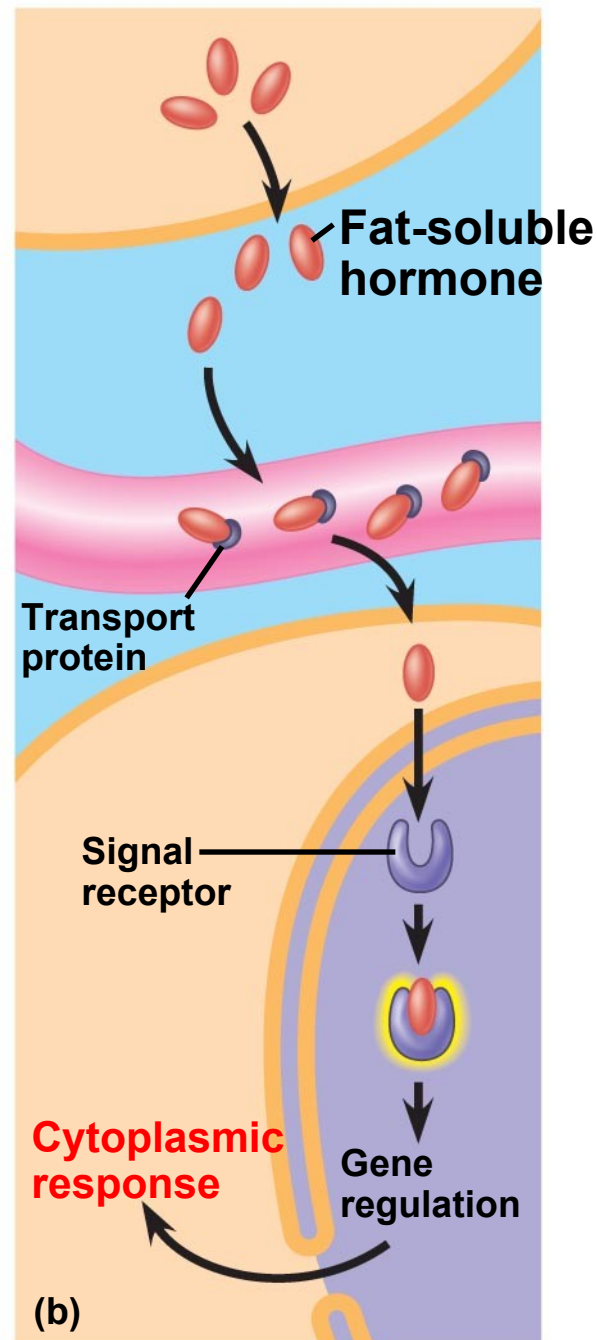
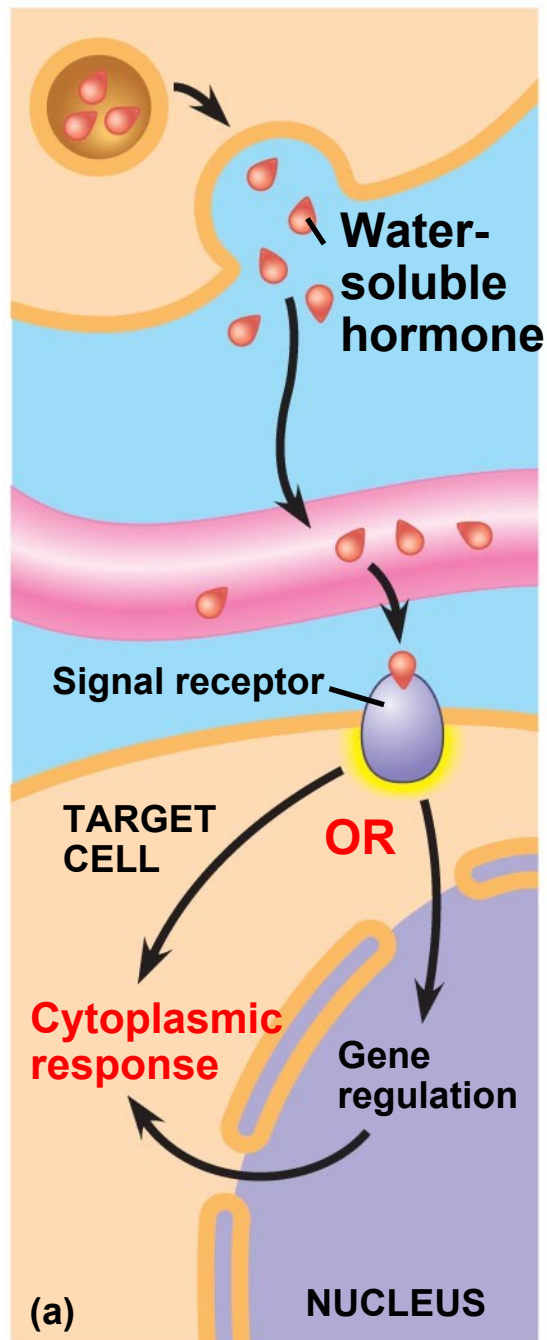
signal transduction pathway

- **Signaling** by any of these hormones involves three key events:
 - **Reception**
 - **Signal transduction**
 - **Response**
- Binding of a hormone to its receptor initiates a **signal transduction pathway** leading to responses in the cytoplasm, enzyme activation, or a change in gene expression.

Receptor location varies with hormone type



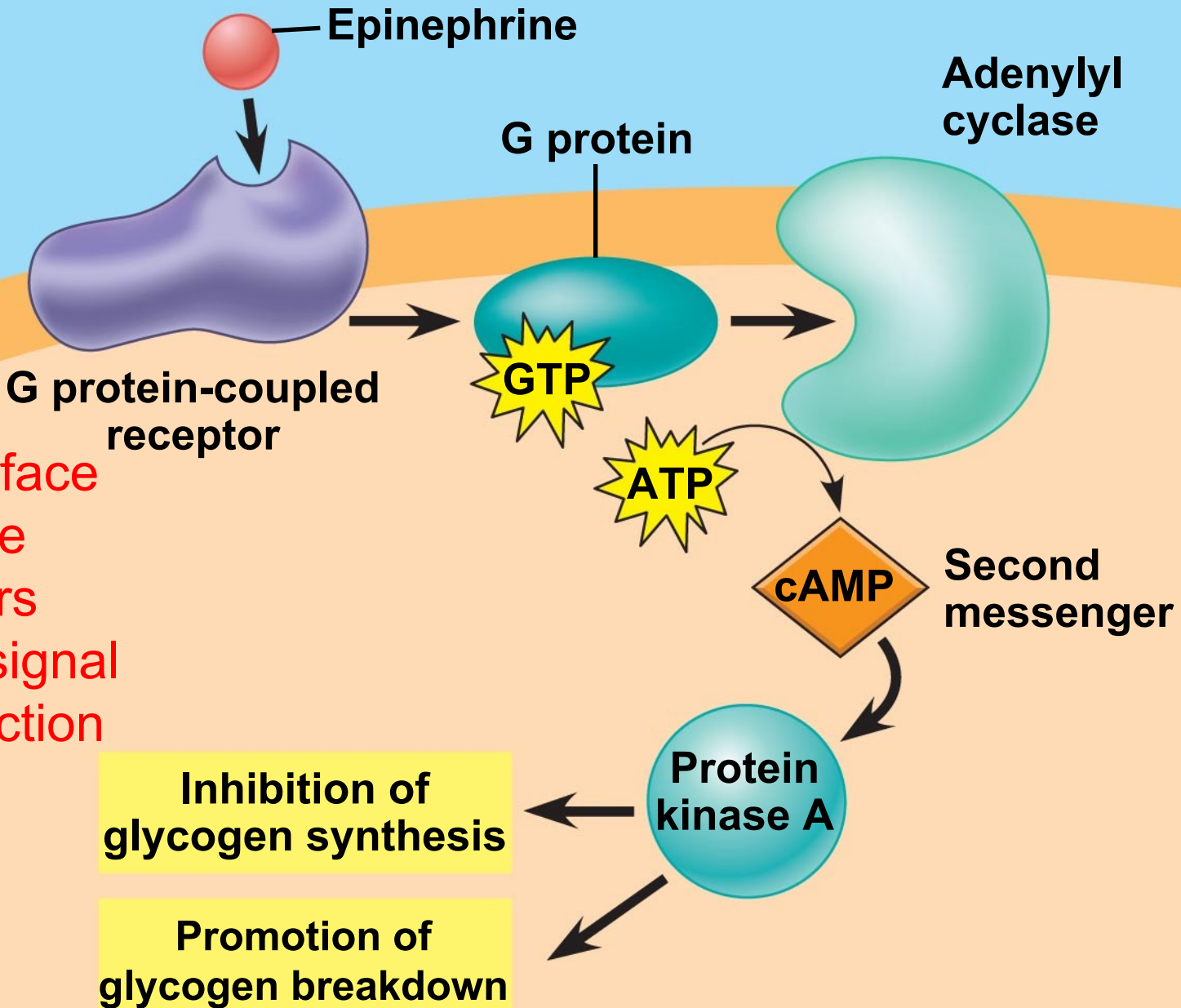
Receptor location varies with hormone type



Pathway for Water-Soluble Hormones

- The hormone **epinephrine** has multiple effects in mediating the body's response to short-term stress.
- Epinephrine *binds to receptors on the plasma membrane* of liver cells.
- This *triggers the release of messenger molecules that activate enzymes* and result in the release of glucose into the bloodstream.

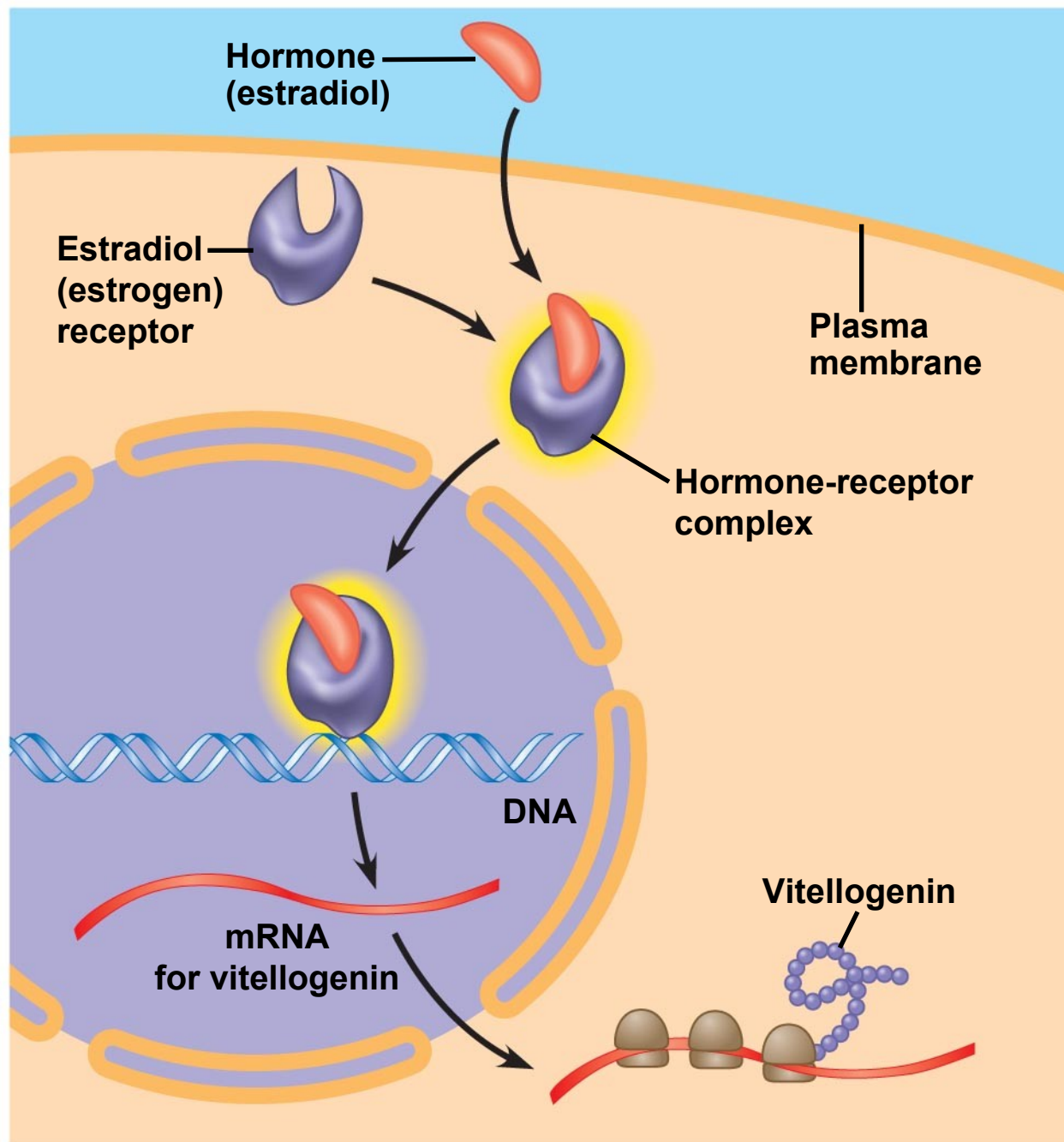
Cell-surface
hormone
receptors
trigger signal
transduction



Pathway for Lipid-Soluble Hormones

- The *response to a lipid-soluble hormone is usually a change in gene expression.*
- Steroids, thyroid hormones, and the hormonal form of vitamin D enter target cells and *bind to protein receptors in the cytoplasm or nucleus.*
- Protein-receptor complexes then act as *transcription factors* in the nucleus, regulating transcription of specific genes.

Steroid hormone receptors are inside the cell and directly regulate gene expression



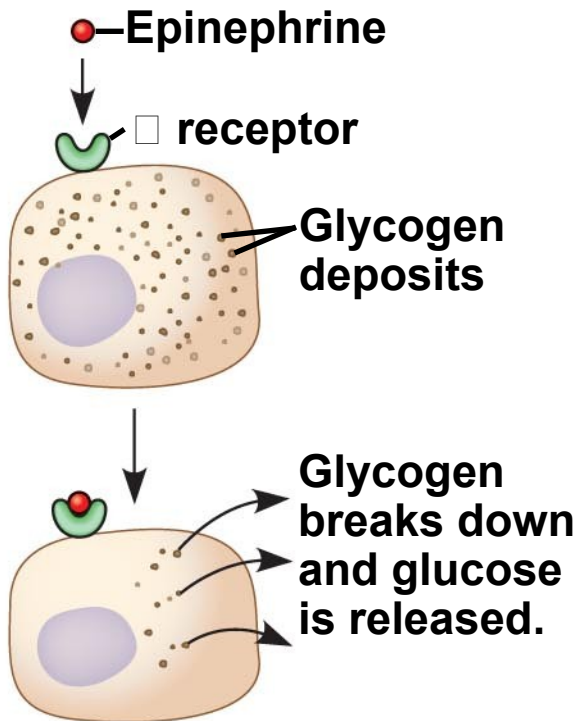
Multiple Effects of Hormones

- The same hormone may have different effects on target cells that have
 - *Different receptors* for the hormone
 - *Different signal transduction pathways*
 - *Different proteins* for carrying out the response.
- A hormone can also have different effects in different species.

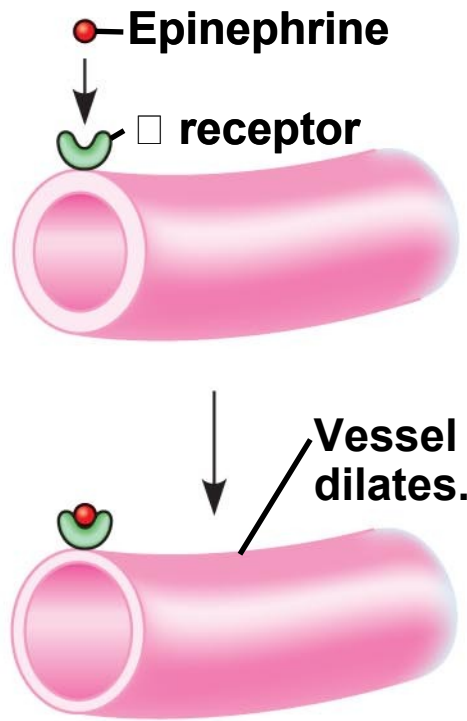
One hormone, different effects

Same receptors but **different intracellular proteins**

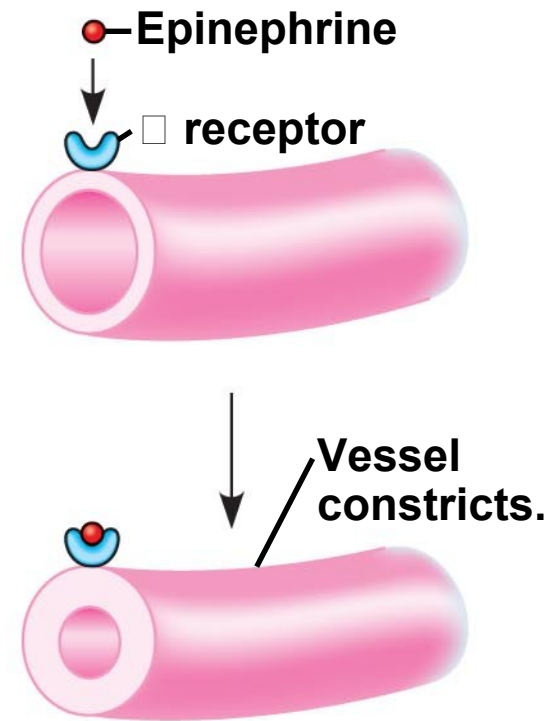
Different receptors



(a) Liver cell

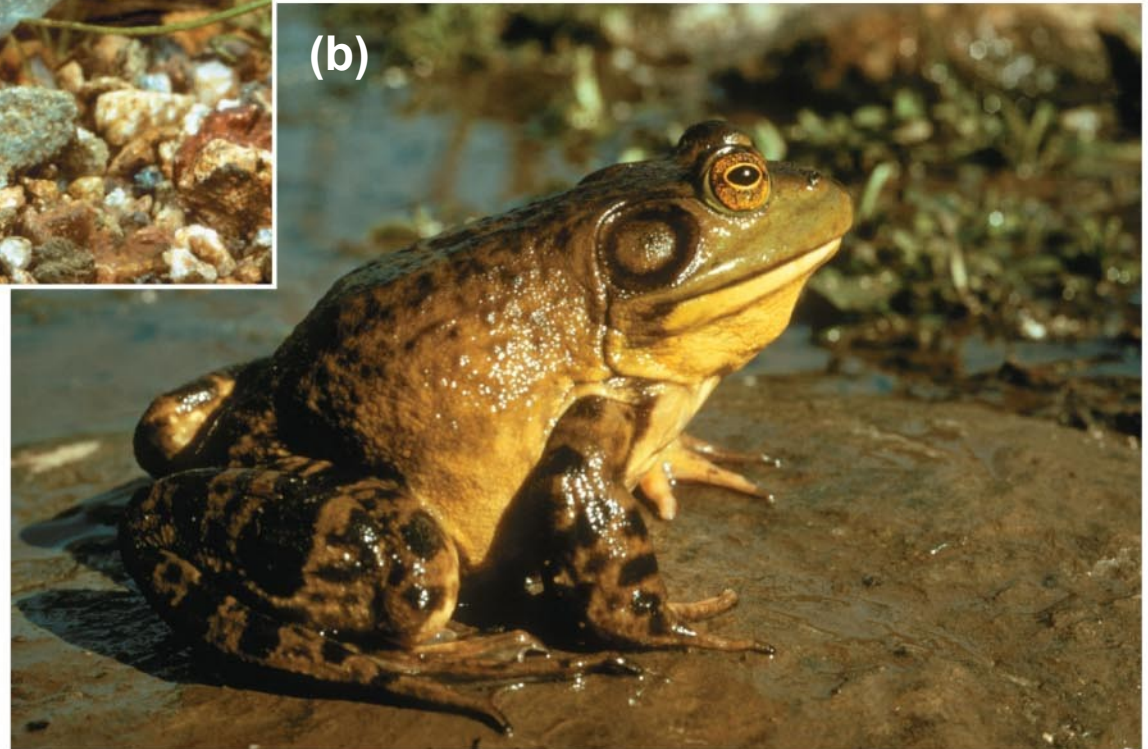


(b) Skeletal muscle blood vessel



(c) Intestinal blood vessel

Specialized role of a hormone in frog metamorphosis



Signaling by **Local Regulators**

- In paracrine signaling, nonhormonal chemical signals called local regulators elicit responses in nearby target cells.
- **Types of local regulators:**
 - **Cytokines** and **growth factors**
 - **Nitric oxide (NO)**
 - **Prostaglandins** - help regulate aggregation of platelets, an early step in formation of blood clots.

Major endocrine glands:

Hypothalamus

Pineal gland

Pituitary gland

Thyroid gland

Parathyroid glands

Adrenal glands

Pancreas

Kidney

Ovaries

Organs containing
endocrine cells:

Thymus

Heart

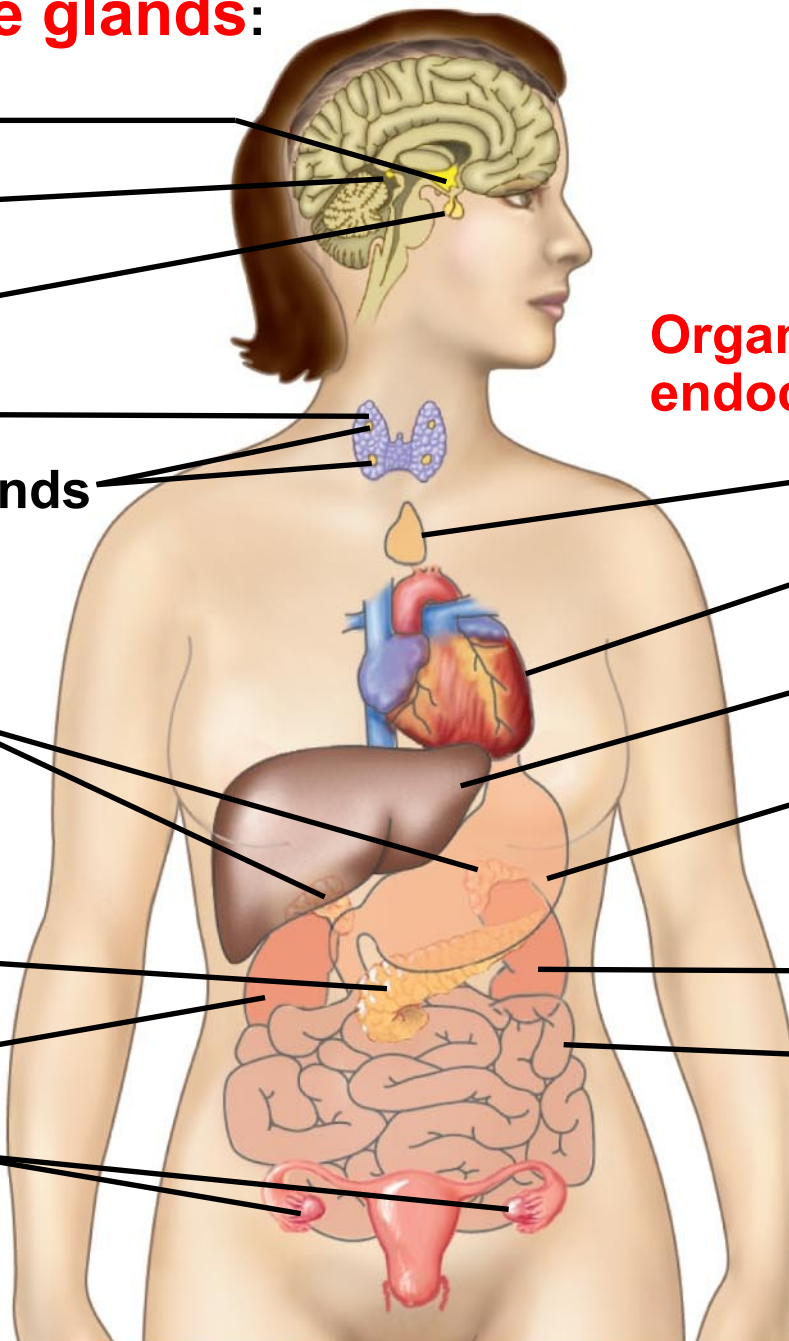
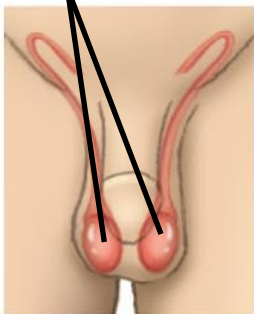
Liver

Stomach

Kidney

Small
intestine

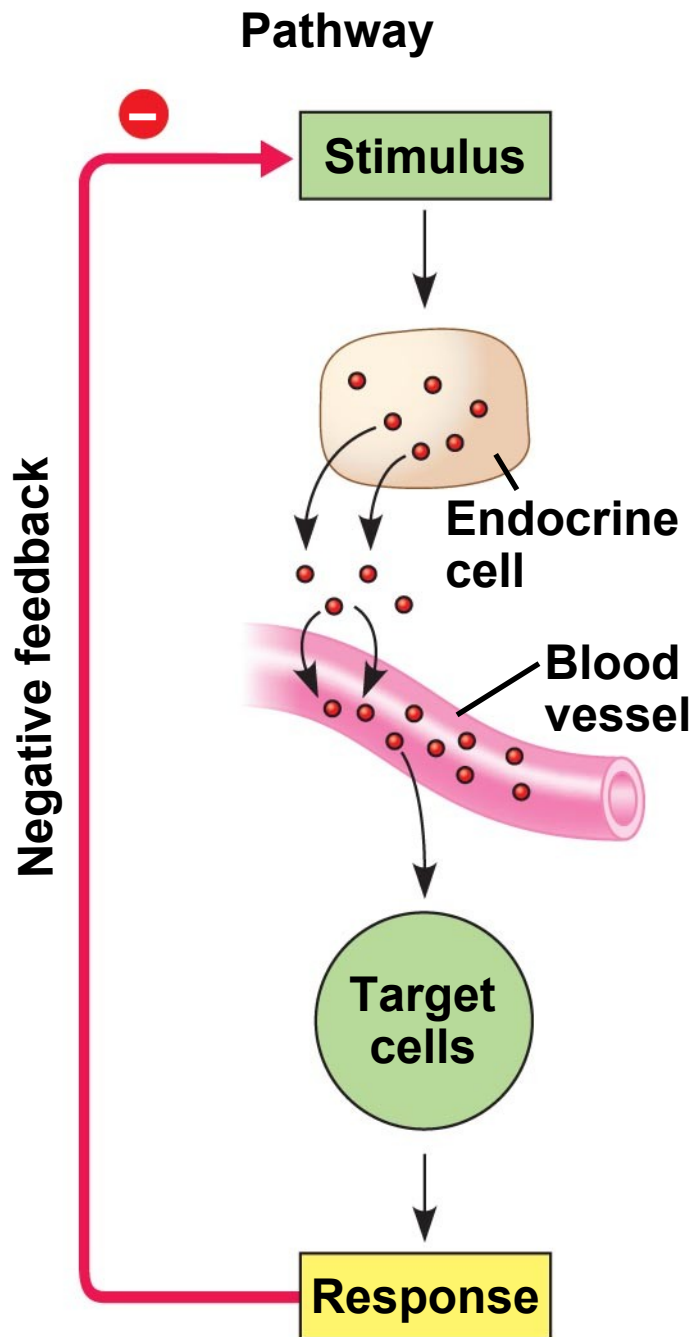
Testes



Simple Hormone Pathways

- **Negative feedback** and **antagonistic hormone pairs** are common features of the endocrine system.
- Hormones are assembled into regulatory pathways.
- ***Hormones*** are released from an ***endocrine cell***, travel through the ***bloodstream***, and interact with the ***receptor*** or a ***target cell*** to cause a ***physiological response***.

A simple endocrine pathway



Example

Low pH in duodenum

S cells of duodenum secrete secretin (●)

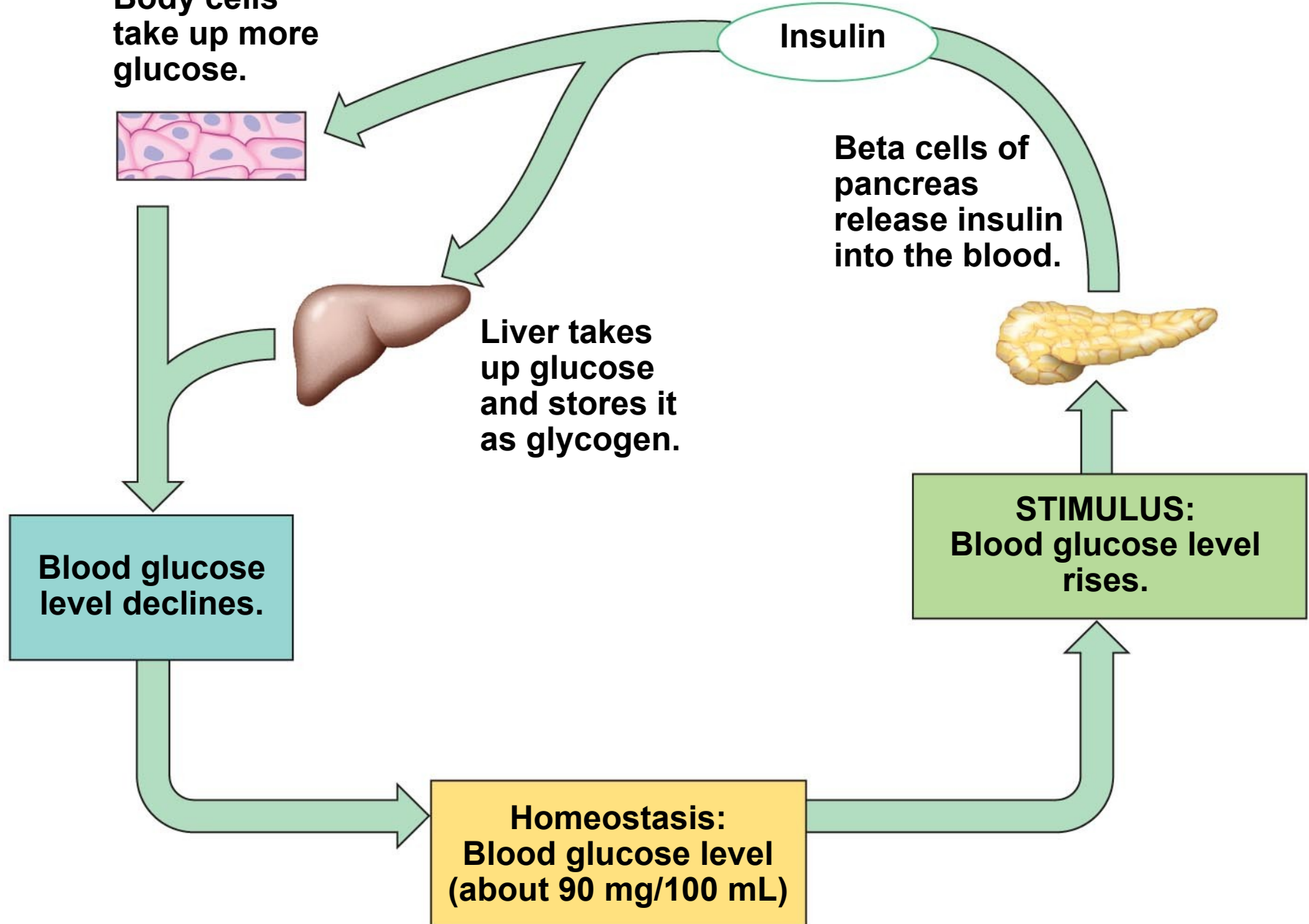
Pancreas

Bicarbonate release

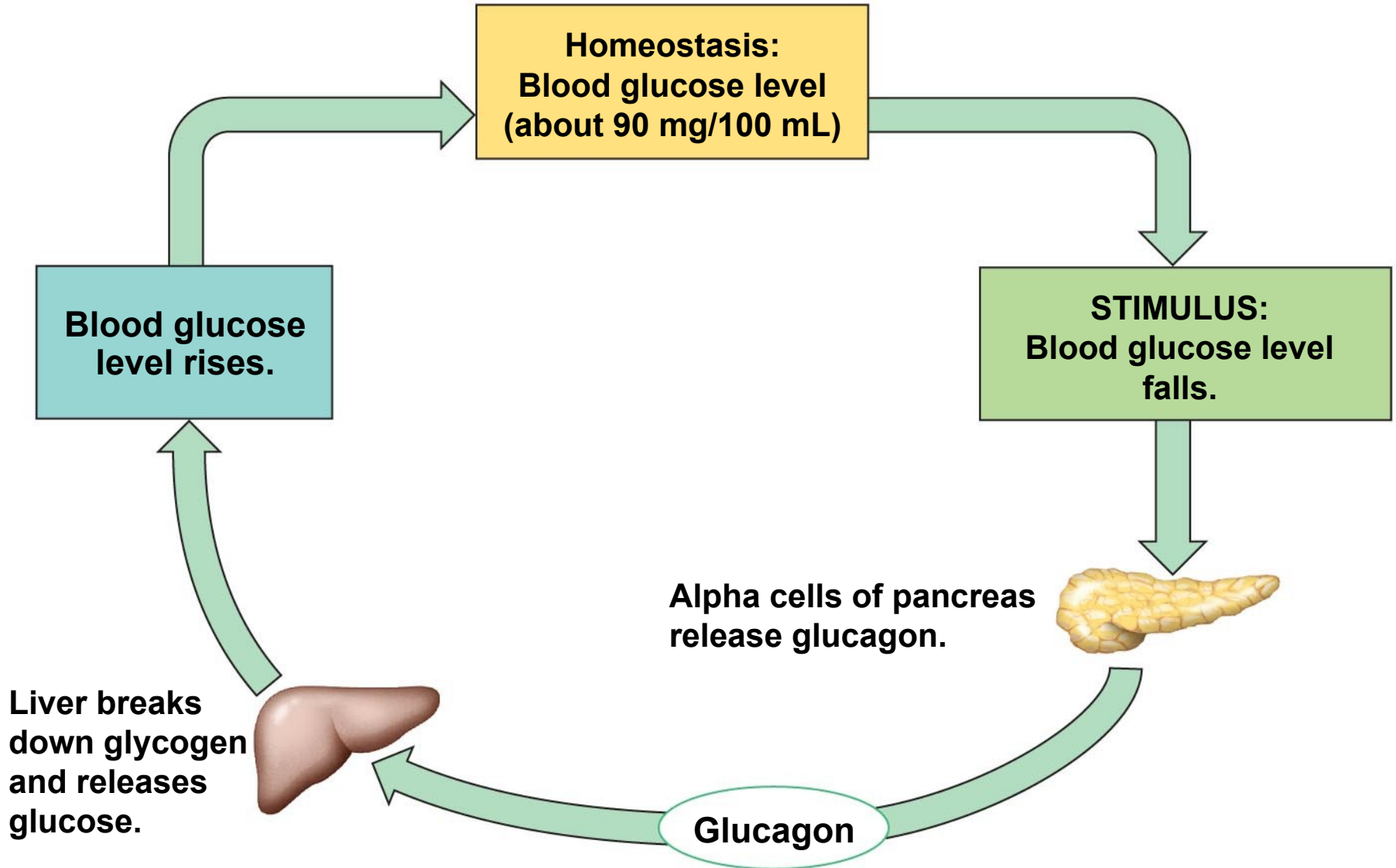
Insulin and Glucagon: Control of Blood Glucose

- A **negative feedback** loop inhibits a response by reducing the initial stimulus.
- Negative feedback reverses a trend to regulate many hormonal pathways involved in **homeostasis**.
- **Insulin** and **glucagon** are **antagonistic hormones** that help maintain glucose homeostasis.
- The **pancreas** has endocrine cells called **islets of Langerhans** with *alpha cells* that produce glucagon and *beta cells* that produce insulin.

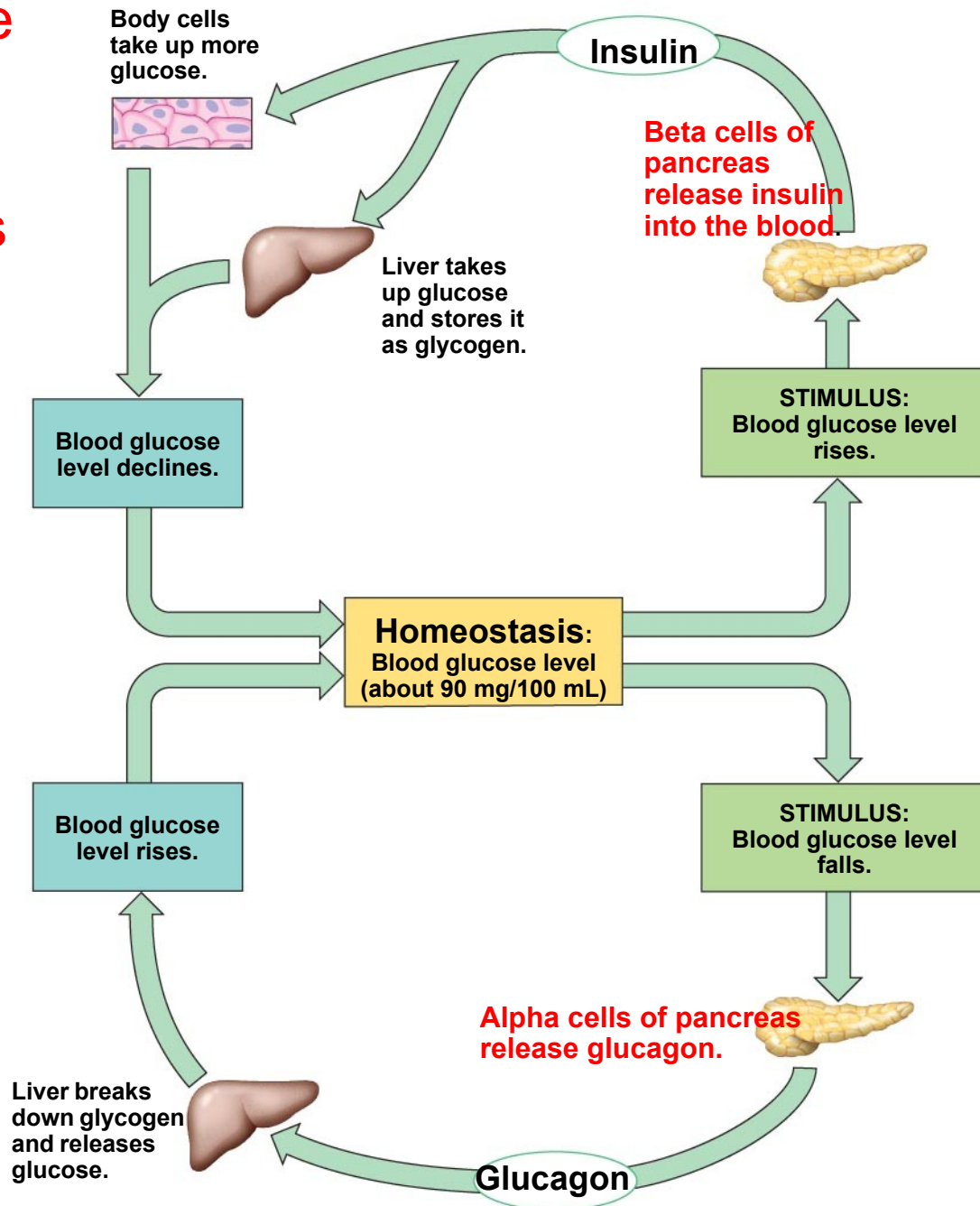
Insulin Lowers Blood Glucose Levels



Glucagon Raises Blood Glucose Levels



Maintenance of glucose homeostasis by insulin and glucagon



Target Tissues for Insulin and Glucagon

- ***Insulin reduces blood glucose levels*** by
 - Promoting the cellular uptake of glucose
 - Slowing glycogen breakdown in the liver
 - Promoting fat storage.
- ***Glucagon increases blood glucose levels*** by
 - Stimulating conversion of glycogen to glucose in the liver
 - Stimulating breakdown of fat and protein into glucose.

Diabetes Mellitus

- **Diabetes mellitus** is an endocrine disorder caused by a deficiency of insulin or a decreased response to insulin in target tissues.
- It is marked by elevated blood glucose levels.
- *Type I diabetes mellitus* (insulin-dependent) is an autoimmune disorder in which the immune system destroys pancreatic beta cells.
- *Type II diabetes mellitus* (non-insulin-dependent) involves insulin deficiency or reduced response of target cells due to change in insulin receptors.

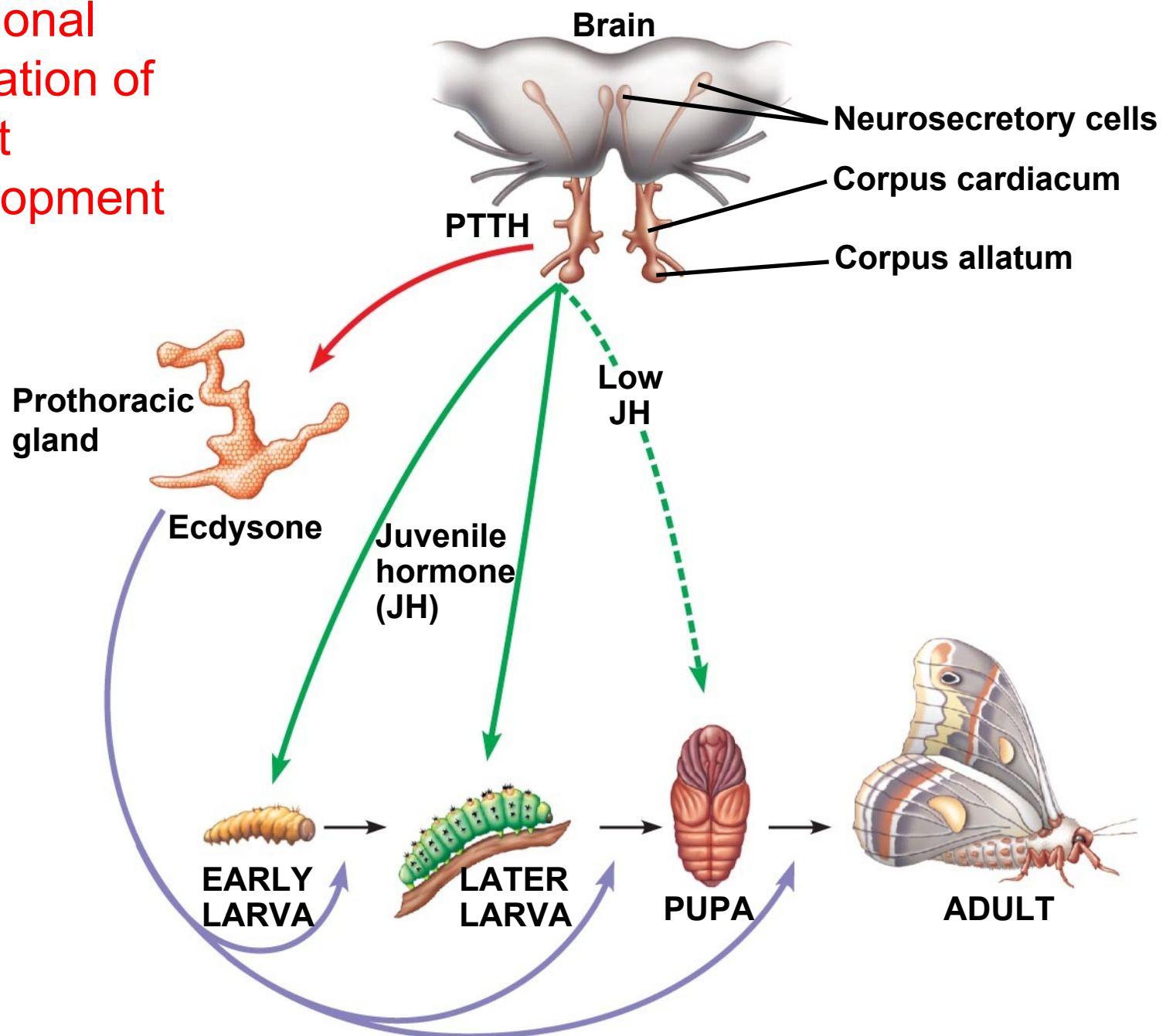
The endocrine and nervous systems act individually and together in regulating animal physiology

- ***Signals from the nervous system initiate and regulate endocrine signals.***

Coordination of Endocrine and Nervous Systems in Invertebrates

- In insects, *molting and development are controlled by a combination of hormones*:
 - A brain hormone stimulates release of **ecdysone** from the prothoracic glands
 - **Juvenile hormone** promotes retention of larval characteristics
 - Ecdysone promotes molting (in the presence of juvenile hormone) and development (in the absence of juvenile hormone) of adult characteristics

Hormonal regulation of insect development



Coordination of Endocrine and Nervous Systems in Vertebrates

- The **hypothalamus** receives information from the nervous system and initiates responses through the endocrine system.
- Attached to the hypothalamus is the **pituitary gland** composed of the posterior pituitary and anterior pituitary.

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- The **posterior pituitary** stores and secretes hormones that are made in the hypothalamus
 - The **anterior pituitary** makes and releases hormones under regulation of the hypothalamus

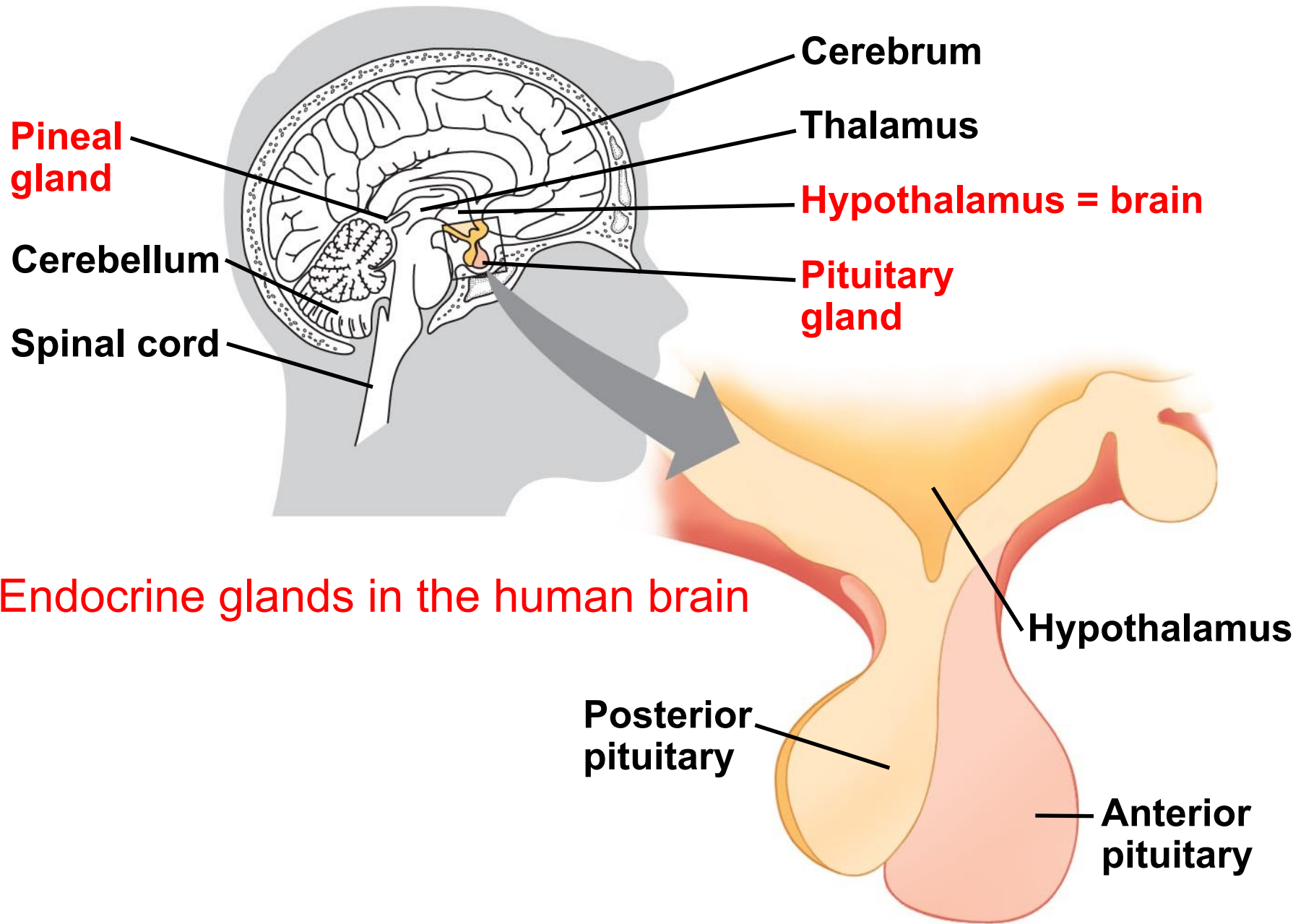


Table 45.1 Major Human Endocrine Glands and Some of Their Hormones

Gland		Hormone	Chemical Class	Representative Actions	Regulated By
Hypothalamus		Hormones released from the posterior pituitary and hormones that regulate the anterior pituitary (see below)			
Posterior pituitary gland (releases neurohormones made in hypothalamus)		Oxytocin	Peptide	Stimulates contraction of uterus and mammary gland cells	Nervous system
		Antidiuretic hormone (ADH)	Peptide	Promotes retention of water by kidneys	Water/salt balance
Anterior pituitary gland		Growth hormone (GH)	Protein	Stimulates growth (especially bones) and metabolic functions	Hypothalamic hormones
		Prolactin (PRL)	Protein	Stimulates milk production and secretion	Hypothalamic hormones
		Follicle-stimulating hormone (FSH)	Glycoprotein	Stimulates production of ova and sperm	Hypothalamic hormones
		Luteinizing hormone (LH)	Glycoprotein	Stimulates ovaries and testes	Hypothalamic hormones
		Thyroid-stimulating hormone (TSH)	Glycoprotein	Stimulates thyroid gland	Hypothalamic hormones
		Adrenocorticotropic hormone (ACTH)	Peptide	Stimulates adrenal cortex to secrete glucocorticoids	Hypothalamic hormones
Thyroid gland		Triiodothyronine (T ₃) and thyroxine (T ₄)	Amine	Stimulate and maintain metabolic processes	TSH
		Calcitonin	Peptide	Lowers blood calcium level	Calcium in blood
Parathyroid glands		Parathyroid hormone (PTH)	Peptide	Raises blood calcium level	Calcium in blood

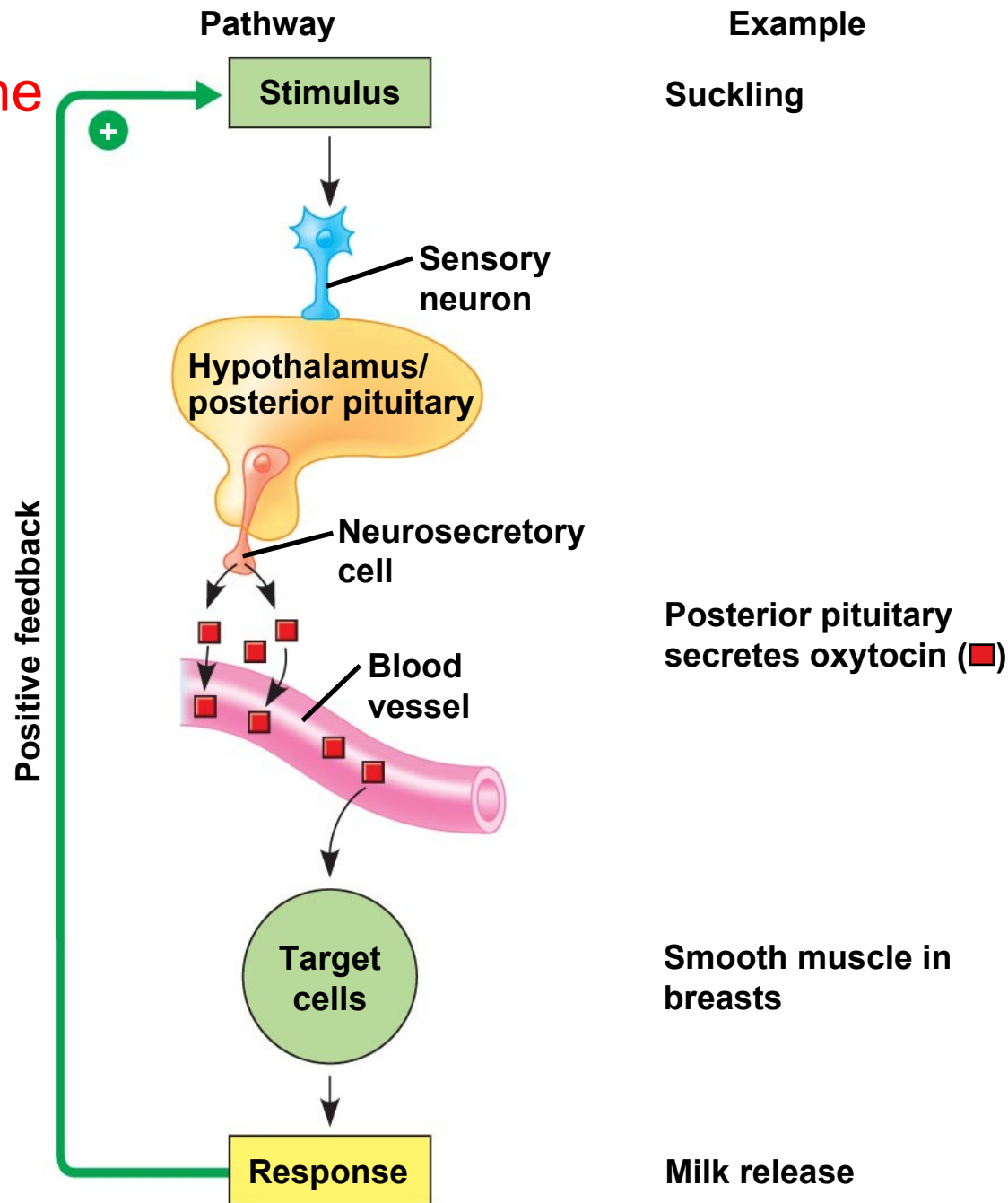
Table 45.1 Major Human Endocrine Glands and Some of Their Hormones

Gland		Hormone	Chemical Class	Representative Actions	Regulated By
Pancreas		Insulin Glucagon	Protein Protein	Lowers blood glucose level Raises blood glucose level	Glucose in blood Glucose in blood
Adrenal glands		Epinephrine and norepinephrine	Amines	Raise blood glucose level; increase metabolic activities; constrict certain blood vessels	Nervous system
Adrenal cortex		Glucocorticoids Mineralocorticoids	Steroid Steroid	Raise blood glucose level Promote reabsorption of Na ⁺ and excretion of K ⁺ in kidneys	ACTH K ⁺ in blood; angiotensin II
Gonads					
Testes		Androgens	Steroid	Support sperm formation; promote development and maintenance of male secondary sex characteristics	FSH and LH
Ovaries		Estrogens	Steroid	Stimulate uterine lining growth; promote development and maintenance of female secondary sex characteristics	FSH and LH
		Progestins	Steroid	Promote uterine lining growth	FSH and LH
Pineal gland		Melatonin	Amine	Involved in biological rhythms	Light/dark cycles

Posterior Pituitary Hormones

- **Oxytocin** induces uterine contractions and the release of milk
- Suckling sends a message to the hypothalamus via the nervous system to release oxytocin, which further stimulates the milk glands
- This is an example of **positive feedback**, where the stimulus leads to an even greater response
- **Antidiuretic hormone (ADH)** enhances water reabsorption in the kidneys

A simple neurohormone pathway



Anterior Pituitary Hormones

- Hormone production in the anterior pituitary is controlled by releasing and inhibiting hormones from the hypothalamus
- For example, the production of *thyrotropin releasing hormone* (**TRH**) in the hypothalamus stimulates secretion of the *thyroid stimulating hormone* (**TSH**) from the anterior pituitary

Production and release of anterior pituitary hormones

Tropic effects only:

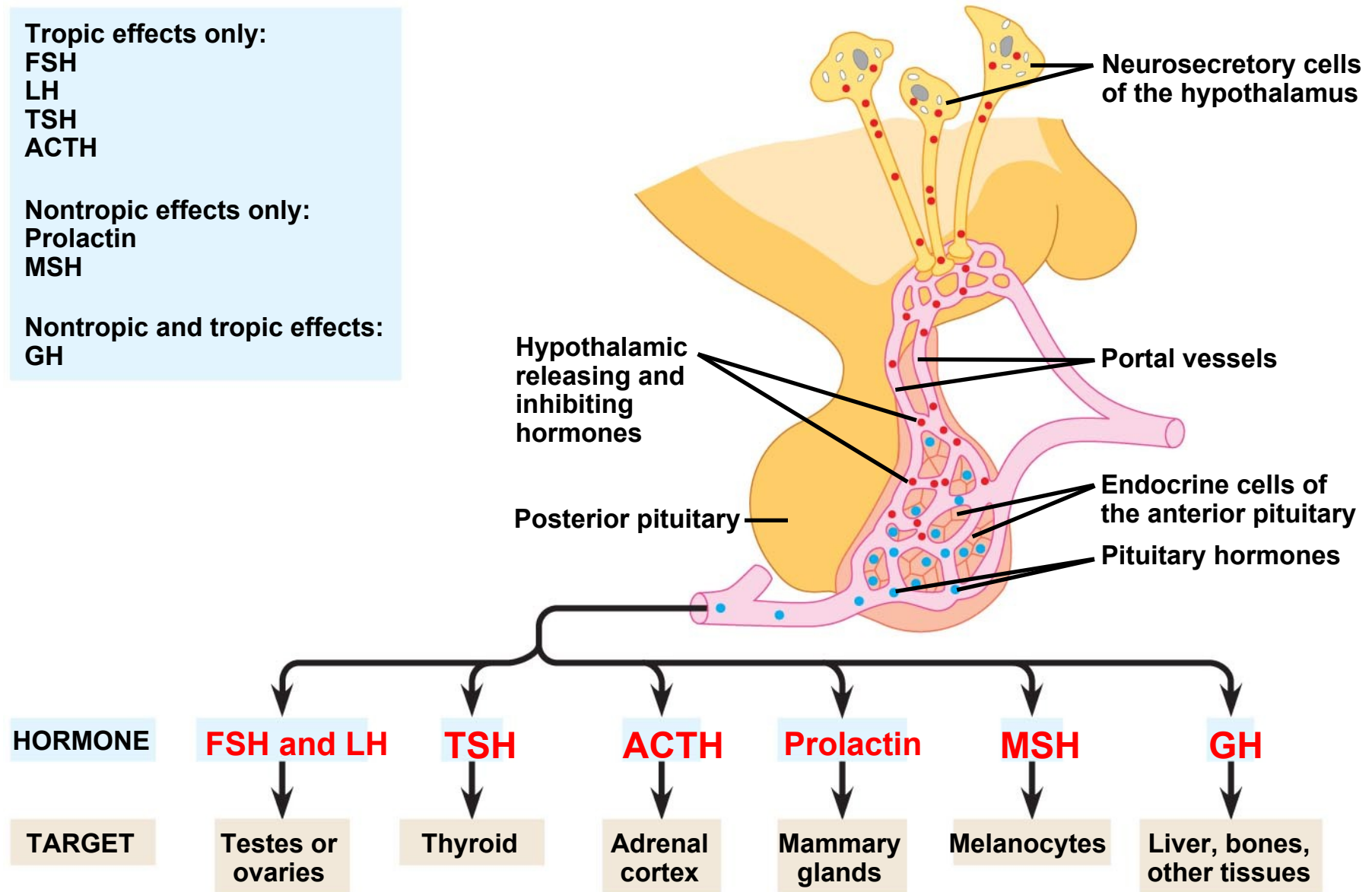
FSH
LH
TSH
ACTH

Nontropic effects only:

Prolactin
MSH

Nontropic and tropic effects:

GH



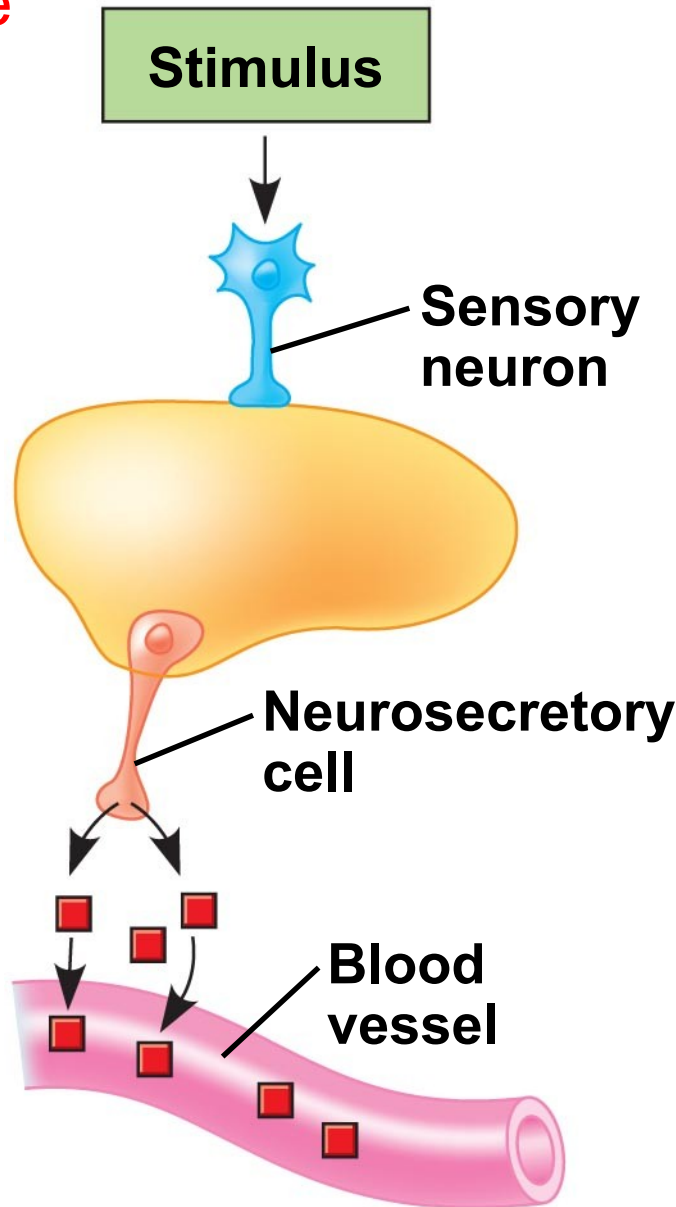
Hormone Cascade Pathways

- A hormone can stimulate the release of a series of other hormones, the last of which activates a nonendocrine target cell; this is called a hormone cascade pathway.
- The release of thyroid hormone results from a hormone cascade pathway involving the hypothalamus, anterior pituitary, and thyroid gland.
- *Hormone cascade pathways are usually regulated by negative feedback.*

A
hormone
cascade
pathway

Pathway

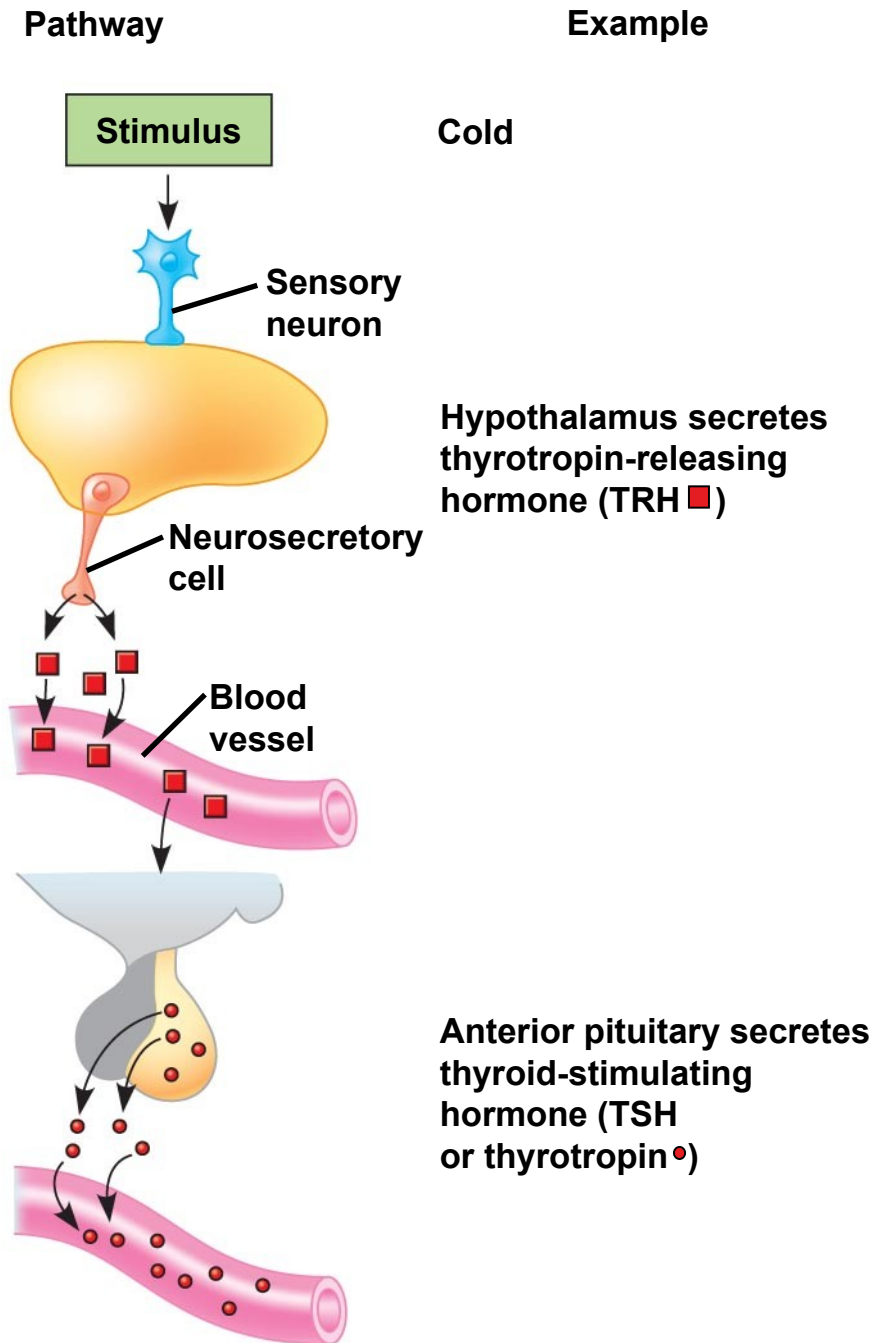
Example



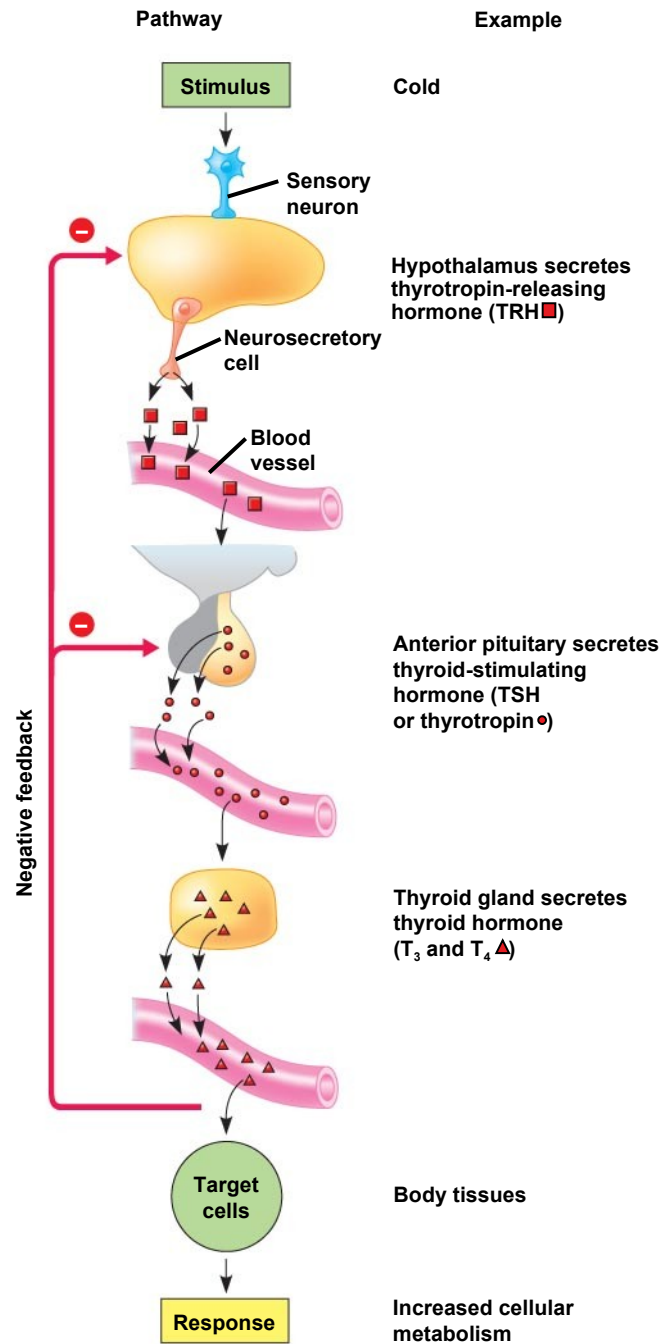
Cold

**Hypothalamus secretes
thyrotropin-releasing
hormone (TRH ■)**

A hormone casade pathway



A hormone cascade pathway



Tropic Hormones

- A **tropic** hormone **regulates** the function of **endocrine** cells or **glands**.
- The four strictly **tropic hormones** are:
 - **Thyroid-stimulating hormone (TSH)**
 - **Follicle-stimulating hormone (FSH)**
 - **Luteinizing hormone (LH)**
 - **Adrenocorticotrophic hormone (ACTH)**

Nontropic Hormones - target **nonendocrine** tissues.

- Nontropic hormones produced by the anterior pituitary are:
 - **Prolactin (PRL)**
 - **Melanocyte-stimulating hormone (MSH)**
- **Prolactin** stimulates **lactation** in mammals but has diverse effects in different vertebrates.
- **MSH** influences **skin pigmentation** in some vertebrates and fat metabolism in mammals.

Growth Hormone

- **Growth hormone (GH)** is secreted by the anterior pituitary gland and has tropic and nontropic actions.
- It promotes growth directly and has diverse metabolic effects.
- It *stimulates production of growth factors*.
- An excess of GH can cause gigantism, while a lack of GH can cause dwarfism.

Endocrine glands respond to diverse stimuli in regulating metabolism, homeostasis, development, and behavior

- Endocrine signaling regulates metabolism, homeostasis, development, and behavior.

Thyroid Hormone: Control of Metabolism and Development

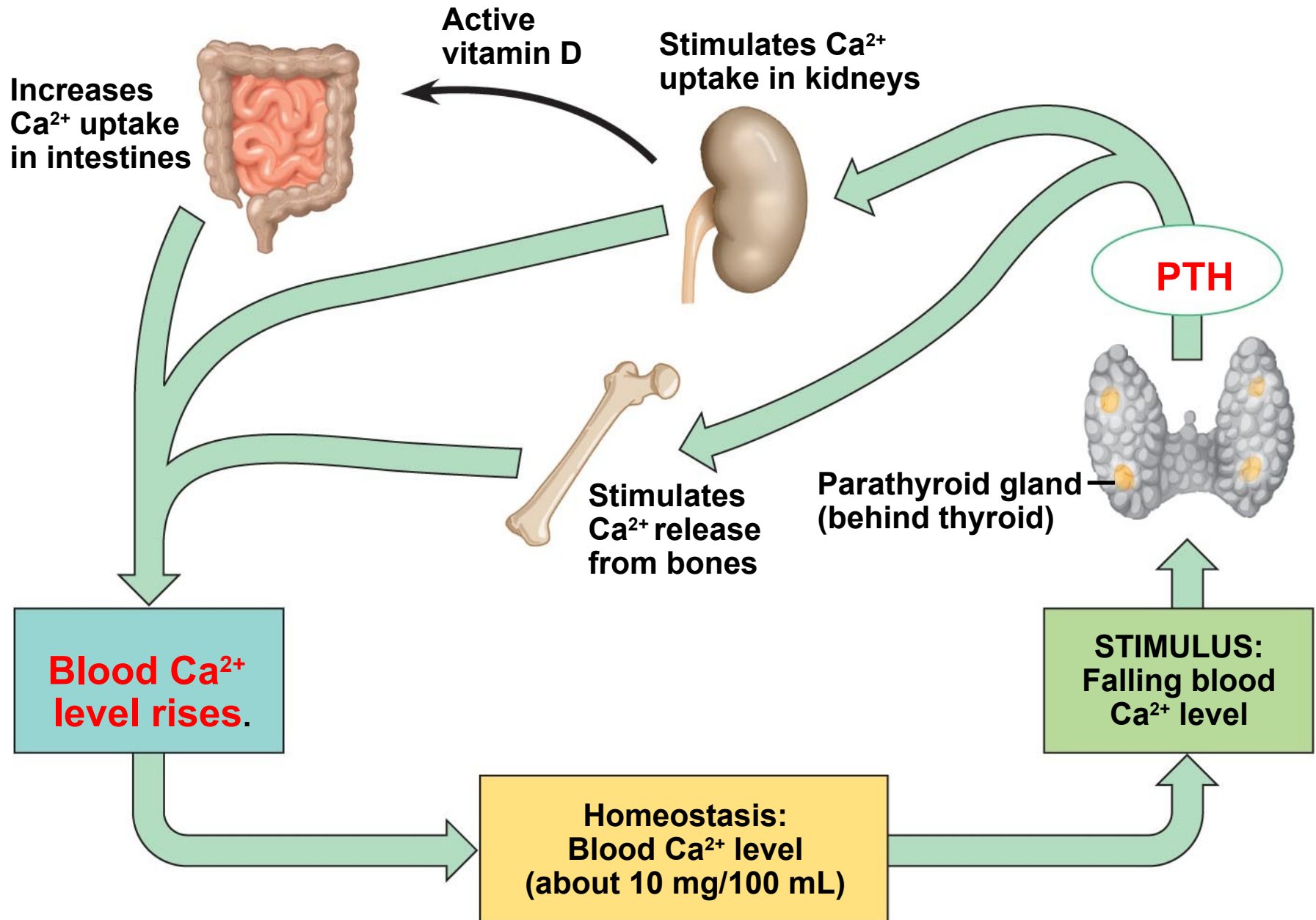
- The **thyroid gland** consists of two lobes on the ventral surface of the trachea.
- It produces two iodine-containing **hormones**: **triiodothyronine (T₃)** and **thyroxine (T₄)**.
- Proper thyroid function requires dietary **iodine** for thyroid hormone production.

-
- **Thyroid hormones stimulate metabolism** and influence development and maturation.
 - **Hyperthyroidism**, excessive secretion of thyroid hormones, causes high body temperature, weight loss, irritability, and high blood pressure.
 - *Graves' disease is a form of hyperthyroidism in humans.*
 - **Hypothyroidism**, low secretion of thyroid hormones, causes weight gain, lethargy, and intolerance to cold.
-

Parathyroid Hormone and Vitamin D: Control of Blood Calcium

- Two **antagonistic hormones** regulate the homeostasis of calcium (Ca^{2+}) in the blood of mammals
 - **Parathyroid hormone (PTH)** is released by the **parathyroid glands**
 - **Calcitonin** is released by the thyroid gland

Antagonistic Hormone Pairs control blood calcium levels



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- **PTH increases the level of blood Ca^{2+}**
 - It releases Ca^{2+} from bone and stimulates reabsorption of Ca^{2+} in the kidneys
 - It also has an indirect effect, stimulating the kidneys to activate vitamin D, which promotes intestinal uptake of Ca^{2+} from food
 - **Calcitonin decreases the level of blood Ca^{2+}**
 - It stimulates Ca^{2+} deposition in bones and secretion by kidneys

Adrenal Hormones: Response to Stress

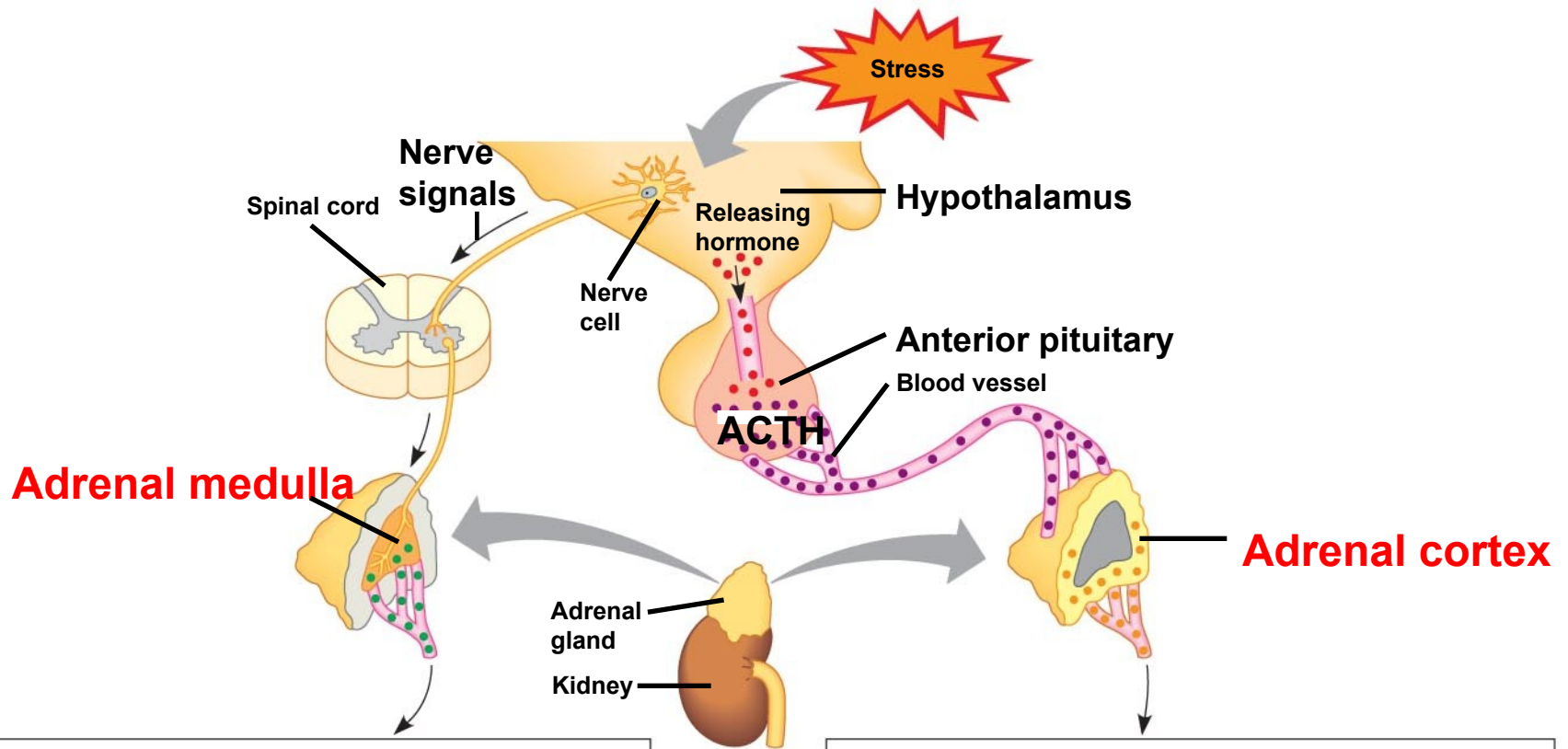
- The *adrenal glands* are adjacent to the kidneys.
- Each adrenal gland actually consists of two glands: the *adrenal medulla* (inner portion) and *adrenal cortex* (outer portion).

Catecholamines from the Adrenal Medulla

- The adrenal medulla secretes *epinephrine* (adrenaline) and *norepinephrine* (noradrenaline).
- These hormones are members of a class of compounds called **catecholamines**.
- They are secreted in response to stress-activated impulses from the nervous system.
- They mediate various **fight-or-flight responses**.

-
- Epinephrine and norepinephrine
 - Trigger the release of glucose and fatty acids into the blood
 - Increase oxygen delivery to body cells
 - Direct blood toward heart, brain, and skeletal muscles, and away from skin, digestive system, and kidneys.
 - *The release of epinephrine and norepinephrine occurs in response to nerve signals from the hypothalamus.*

Summary: Stress and the Adrenal Gland



(a) Short-term stress response

Effects of **epinephrine** and **norepinephrine**:

1. Glycogen broken down to glucose; increased blood glucose
2. Increased blood pressure
3. Increased breathing rate
4. Increased metabolic rate
5. Change in blood flow patterns, leading to increased alertness and decreased digestive, excretory, and reproductive system activity

(b) Long-term stress response

Effects of

mineralocorticoids:

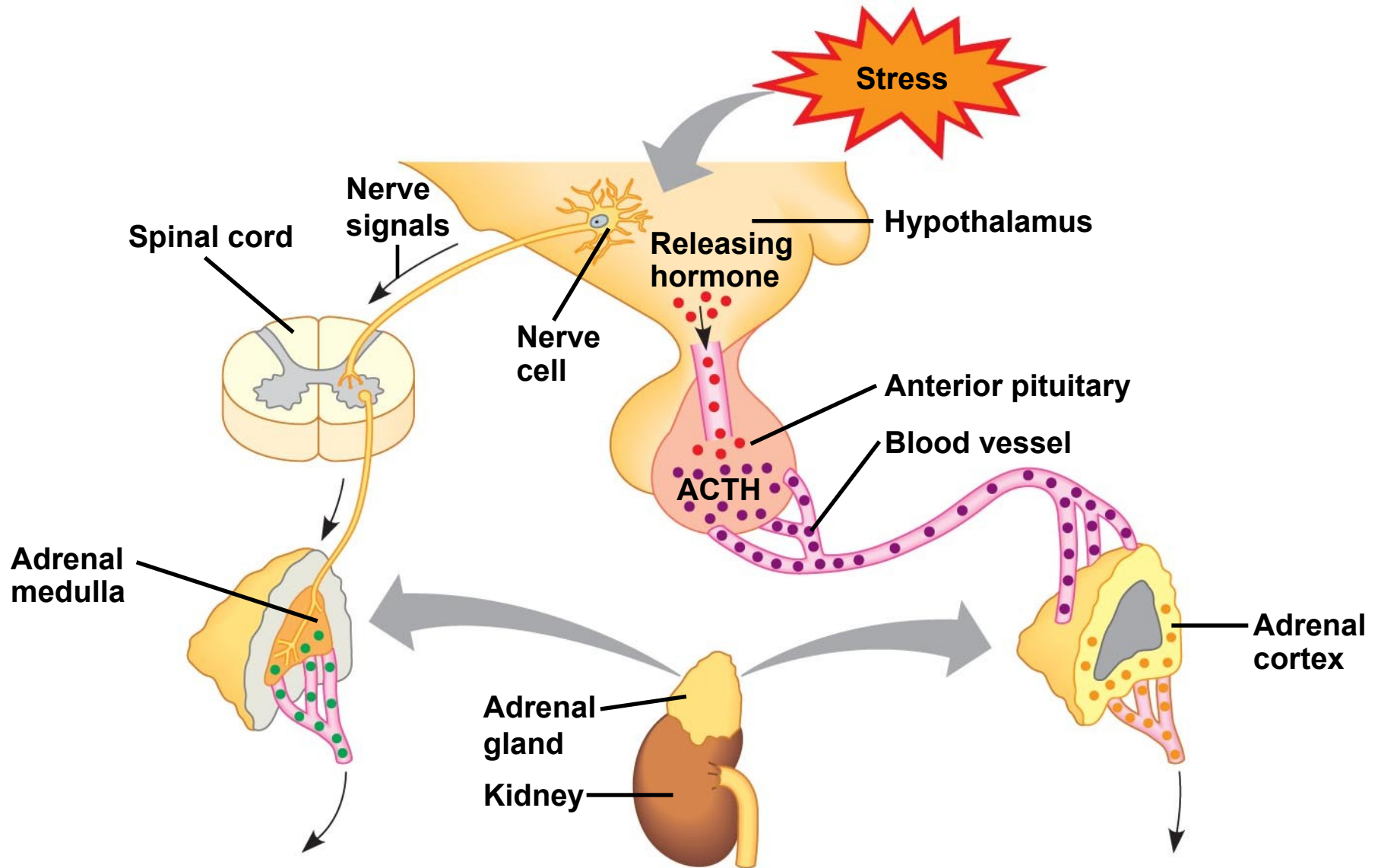
1. Retention of sodium ions and water by kidneys
2. Increased blood volume and blood pressure

Effects of

glucocorticoids:

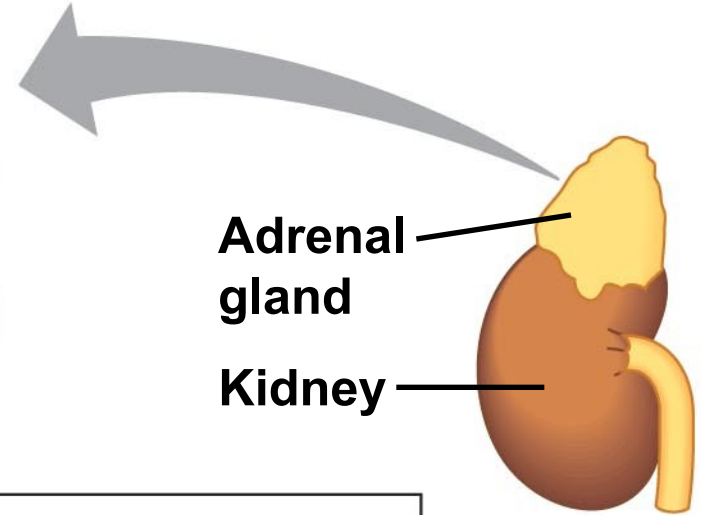
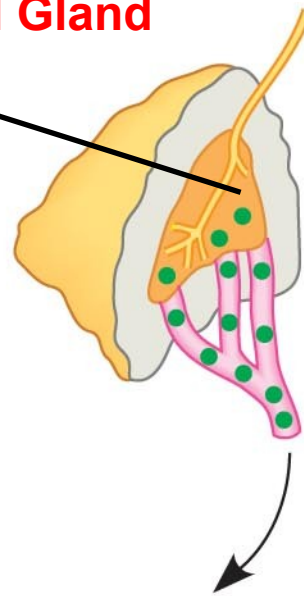
1. Proteins and fats broken down and converted to glucose, leading to increased blood glucose
2. Possible suppression of immune system

Stress and the Adrenal Gland



Short-term Stress and the Adrenal Gland

Adrenal medulla



(a) Short-term stress response

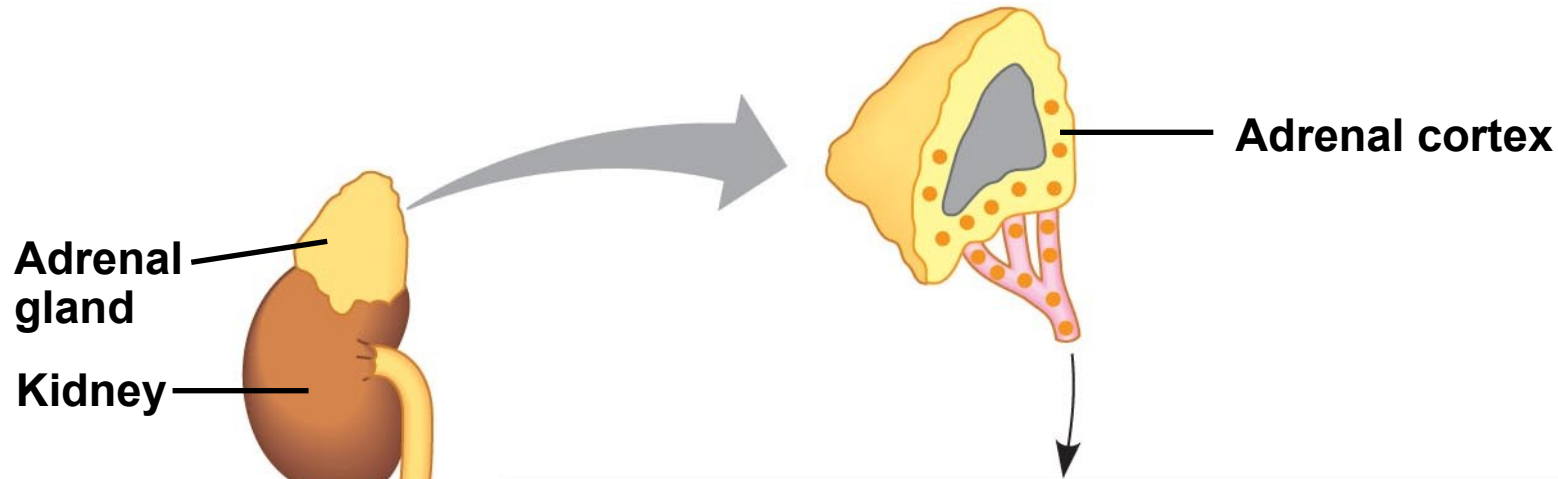
Effects of **epinephrine** and **norepinephrine**:

1. Glycogen broken down to glucose; increased blood glucose
2. Increased blood pressure
3. Increased breathing rate
4. Increased metabolic rate
5. Change in blood flow patterns, leading to increased alertness and decreased digestive, excretory, and reproductive system activity

Steroid Hormones from the Adrenal Cortex

- The **adrenal cortex** releases a family of steroids called **corticosteroids** in response to **stress**.
- These hormones are triggered by a hormone cascade pathway via the hypothalamus and anterior pituitary.
- Humans produce two types of corticosteroids: glucocorticoids and mineralocorticoids.

Long-term Stress and the adrenal gland



(b) Long-term stress response

Effects of **mineralocorticoids:**

1. Retention of sodium ions and water by kidneys
2. Increased blood volume and blood pressure

Effects of **glucocorticoids:**

1. Proteins and fats broken down and converted to glucose, leading to increased blood glucose
2. Possible suppression of immune system

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- **Glucocorticoids**, such as cortisol, influence **glucose** metabolism and the immune system.
 - **Mineralocorticoids**, such as aldosterone, affect **salt** and **water** balance.
 - The adrenal cortex also produces small amounts of steroid hormones that function as sex hormones.

Gonadal Sex Hormones

- The **gonads** = **testes and ovaries**, produce most of the **sex hormones**: androgens, estrogens, and progestins.
- All three sex hormones are found in both males and females, but in different amounts.

-
- The **testes** primarily synthesize **androgens**, mainly **testosterone**, which stimulate development and maintenance of the **male reproductive system** and **male secondary sex characteristics**.
 - Testosterone causes an increase in muscle and bone mass and is often taken as a supplement to cause muscle growth, which carries health risks.

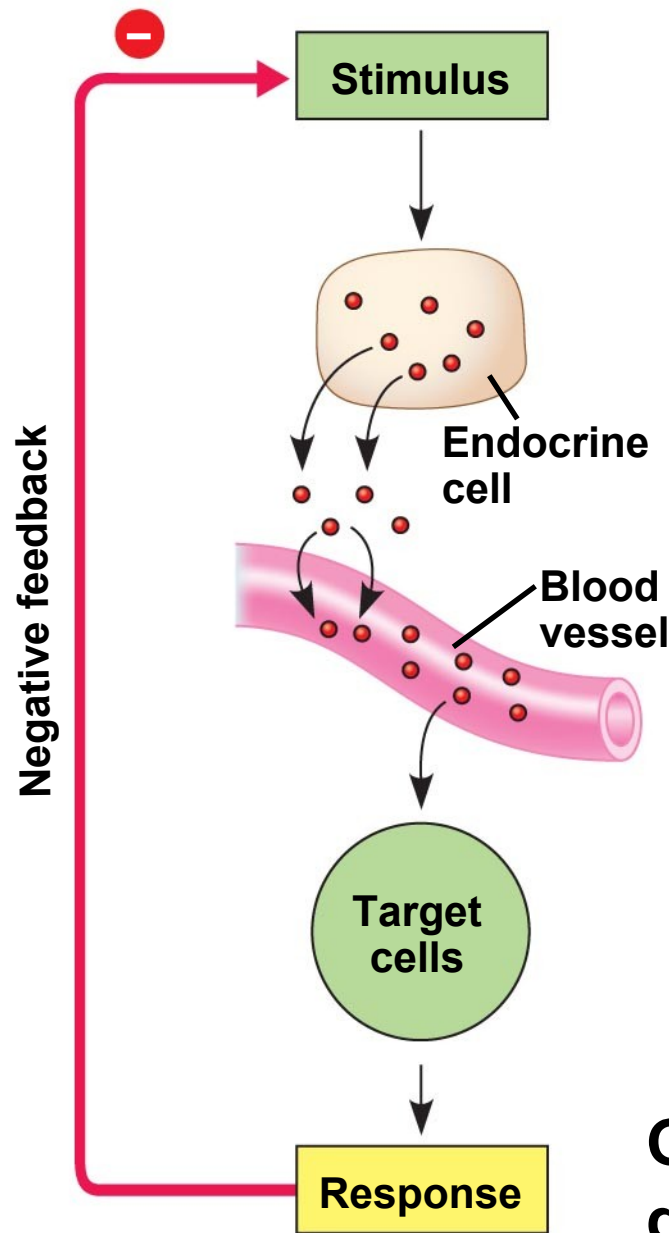
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- **Estrogens**, made in the **ovary**, most importantly **estradiol**, are responsible for maintenance of the **female reproductive system** and the development of **female secondary sex characteristics**.
 - In mammals, progestins, which include **progesterone**, are primarily involved in **preparing and maintaining the uterus**.
 - Synthesis of the sex hormones is controlled by **FSH** and **LH** from the **anterior pituitary**.

Pineal Gland - Melatonin and Biorhythms

- The **pineal gland**, located in the brain, secretes **melatonin**.
- Light/dark cycles control release of melatonin.
- Primary functions of melatonin appear to relate to *biological rhythms associated with reproduction*.

Signal Transduction Pathway

Example



Low blood glucose

**Pancreas alpha cells
secrete[•]
glucagon**

Liver

**Glycogen breakdown,
glucose release into blood**

You should now be able to:

1. Distinguish between the following pairs of terms: hormones and local regulators, paracrine and autocrine signals.
 2. Describe the evidence that steroid hormones have intracellular receptors, while water-soluble hormones have cell-surface receptors.
 3. Explain how the **antagonistic hormones** insulin and glucagon regulate carbohydrate metabolism.
 4. Distinguish between type 1 and type 2 diabetes.
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5. Explain how the hypothalamus and the pituitary glands interact and how they coordinate the endocrine system.
 6. Explain the role of tropic hormones in coordinating endocrine signaling throughout the body.
 7. List and describe the functions of hormones released by the following: anterior and posterior pituitary lobes, thyroid glands, parathyroid glands, adrenal medulla, adrenal cortex, gonads, pineal gland.